

BOILING AND CONDENSATION

We know from thermodynamics that when the temperature of a liquid at a specified pressure is raised to the saturation temperature T_{sat} at that pressure, *boiling* occurs. Likewise, when the temperature of a vapor is lowered to T_{sat} , *condensation* occurs. In this chapter we study the rates of heat transfer during such liquid-to-vapor and vapor-to-liquid phase transformations.

Although boiling and condensation exhibit some unique features, they are considered to be forms of *convection* heat transfer since they involve fluid motion (such as the rise of the bubbles to the top and the flow of condensate to the bottom). Boiling and condensation differ from other forms of convection in that they depend on the *latent heat of vaporization* h_{fg} of the fluid and the *surface tension* σ at the liquid–vapor interface, in addition to the properties of the fluid in each phase. Noting that under equilibrium conditions the temperature remains constant during a phase-change process at a fixed pressure, large amounts of heat (due to the large latent heat of vaporization released or absorbed) can be transferred during boiling and condensation essentially at constant temperature. In practice, however, it is necessary to maintain some difference between the surface temperature T_s and T_{sat} for effective heat transfer. Heat transfer coefficients h associated with boiling and condensation are typically much higher than those encountered in other forms of convection processes that involve a single phase.

We start this chapter with a discussion of the *boiling curve* and the modes of pool boiling such as *free convection boiling*, *nucleate boiling*, and *film boiling*. We then discuss boiling in the presence of forced convection. In the second part of this chapter, we describe the physical mechanism of *film condensation* and discuss condensation heat transfer in several geometrical arrangements and orientations. Finally, we introduce *dropwise condensation* and discuss ways of maintaining it.



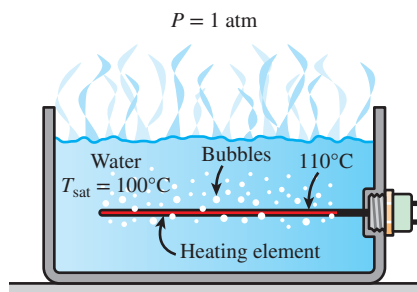
OBJECTIVES

When you finish studying this chapter, you should be able to:

- Differentiate between evaporation and boiling, and gain familiarity with different types of boiling,
- Develop a good understanding of the boiling curve and the different boiling regimes corresponding to different regions of the boiling curve,
- Calculate the heat flux and its critical value associated with nucleate boiling, and examine the methods of boiling heat transfer enhancement,
- Derive a relation for the heat transfer coefficient in laminar film condensation over a vertical plate,
- Calculate the heat flux associated with condensation on inclined and horizontal plates, vertical and horizontal cylinders or spheres, and tube bundles,
- Examine dropwise condensation, and understand the uncertainties associated with them.

**FIGURE 10-1**

A liquid-to-vapor phase change process is called *evaporation* if it originates at a liquid–vapor interface and *boiling* if it occurs at a solid–liquid interface.

**FIGURE 10-2**

Boiling occurs when a liquid is brought into contact with a surface at a temperature above the saturation temperature of the liquid.

10-1 ■ BOILING HEAT TRANSFER

Many familiar engineering applications involve condensation and boiling heat transfer. In a household refrigerator, for example, the refrigerant absorbs heat from the refrigerated space by boiling in the *evaporator* section and rejects heat to the kitchen air by condensing in the *condenser* section (the long coils behind or under the refrigerator). Also, in steam power plants, heat is transferred to the steam in the *boiler* where water is vaporized, and the waste heat is rejected from the steam in the *condenser* where the steam is condensed. Some electronic components are cooled by boiling by immersing them in a fluid with an appropriate boiling temperature.

Boiling is a liquid-to-vapor phase change process just like evaporation, but there are significant differences between the two. **Evaporation** occurs at the *liquid–vapor interface* when the vapor pressure is less than the saturation pressure of the liquid at a given temperature. Water in a lake at 20°C, for example, evaporates to air at 20°C and 60 percent relative humidity since the saturation pressure of water at 20°C is 2.3 kPa and the vapor pressure of air at 20°C and 60 percent relative humidity is 1.4 kPa (evaporation rates are determined in Chap. 14). Other examples of evaporation are the drying of clothes, fruits, and vegetables; the evaporation of sweat to cool the human body; and the rejection of waste heat in wet cooling towers. Note that evaporation involves no bubble formation or bubble motion (Fig. 10-1).

Boiling, on the other hand, occurs at the *solid–liquid interface* when a liquid is brought into contact with a surface maintained at a temperature T_s sufficiently above the saturation temperature T_{sat} of the liquid (Fig. 10-2). At 1 atm, for example, liquid water in contact with a solid surface at 110°C boils since the saturation temperature of water at 1 atm is 100°C. The boiling process is characterized by the rapid formation of *vapor bubbles* at the solid–liquid interface that detach from the surface when they reach a certain size and attempt to rise to the free surface of the liquid. When cooking, we do not say water is boiling until we see the bubbles rising to the top. Boiling is a complicated phenomenon because of the large number of variables involved in the process and the complex fluid motion patterns caused by the bubble formation and growth.

As a form of convection heat transfer, the *boiling heat flux* from a solid surface to the fluid is expressed from Newton's law of cooling as

$$\dot{q}_{\text{boiling}} = h(T_s - T_{\text{sat}}) = h\Delta T_{\text{excess}} \quad (\text{W/m}^2) \quad (10-1)$$

where $\Delta T_{\text{excess}} = T_s - T_{\text{sat}}$ is called the *excess temperature*, which represents the temperature excess of the surface above the saturation temperature of the fluid.

In the preceding chapters we considered forced and free convection heat transfer involving a single phase of a fluid. The analysis of such convection processes involves the thermophysical properties ρ , μ , k , and c_p of the fluid. The analysis of boiling heat transfer involves these properties of the liquid (indicated by the subscript l) or vapor (indicated by the subscript v) as well as the properties h_{fg} (the latent heat of vaporization) and σ (the surface tension). The h_{fg} represents the energy absorbed as a unit mass of liquid vaporizes at a specified temperature or pressure and is the primary quantity of energy transferred during boiling heat transfer. The h_{fg} values of water at various temperatures are given in Table A-9.

Bubbles owe their existence to the *surface-tension* σ at the liquid–vapor interface due to the attraction force on molecules at the interface toward the liquid phase. The surface tension decreases with increasing temperature and becomes zero at the critical temperature. This explains why no bubbles are formed during boiling at supercritical pressures and temperatures. Surface tension has the unit N/m.

The boiling processes in practice do not occur under *equilibrium* conditions, and normally the bubbles are not in thermodynamic equilibrium with the surrounding liquid. That is, the temperature and pressure of the vapor in a bubble are usually different than those of the liquid. The pressure difference between the liquid and the vapor is balanced by the surface tension at the interface. The temperature difference between the vapor in a bubble and the surrounding liquid is the driving force for heat transfer between the two phases. When the liquid is at a *lower temperature* than the bubble, heat is transferred from the bubble into the liquid, causing some of the vapor inside the bubble to condense and the bubble eventually to collapse. When the liquid is at a *higher temperature* than the bubble, heat is transferred from the liquid to the bubble, causing the bubble to grow and rise to the top under the influence of buoyancy.

Boiling is classified as *pool boiling* or *flow boiling*, depending on the presence of bulk fluid motion (Fig. 10–3). Boiling is called **pool boiling** in the absence of bulk fluid flow and **flow boiling** (or *forced convection boiling*) in the presence of it. In pool boiling, the fluid body is stationary, and any motion of the fluid is due to natural convection currents and the motion of the bubbles under the influence of buoyancy. The boiling of water in a pan on top of a stove is an example of pool boiling. Pool boiling of a fluid can also be achieved by placing a heating coil in the fluid. In flow boiling, the fluid is forced to move in a heated pipe or over a surface by external means such as a pump. Therefore, flow boiling is always accompanied by other convection effects.

Pool and flow boiling are further classified as *subcooled boiling* or *saturated boiling*, depending on the bulk liquid temperature (Fig. 10–4). Boiling is said to be **subcooled** (or *local*) when the temperature of the main body of the liquid is below the saturation temperature T_{sat} (i.e., the bulk of the liquid is subcooled) and **saturated** (or *bulk*) when the temperature of the liquid is equal to T_{sat} (i.e., the bulk of the liquid is saturated). At the early stages of boiling, the bubbles are confined to a narrow region near the hot surface. This is because the liquid adjacent to the hot surface vaporizes as a result of being heated above its saturation temperature. But these bubbles disappear soon after they move away from the hot surface as a result of heat transfer from the bubbles to the cooler liquid surrounding them. This happens when the bulk of the liquid is at a lower temperature than the saturation temperature. The bubbles serve as “energy movers” from the hot surface into the liquid body by absorbing heat from the hot surface and releasing it into the liquid as they condense and collapse. Boiling in this case is confined to a region in the locality of the hot surface and is appropriately called *local* or *subcooled* boiling. When the entire liquid body reaches the saturation temperature, the bubbles start rising to the top. We can see bubbles throughout the bulk of the liquid, and boiling in this case is called the *bulk* or *saturated* boiling. Next, we consider different boiling regimes in detail.

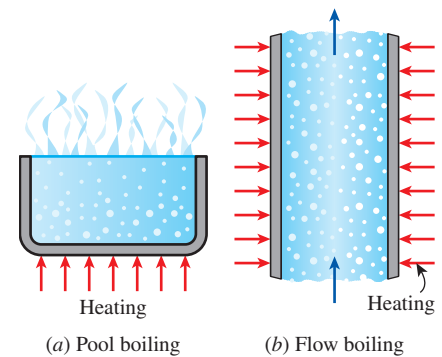


FIGURE 10–3

Classification of boiling on the basis of the presence of bulk fluid motion.

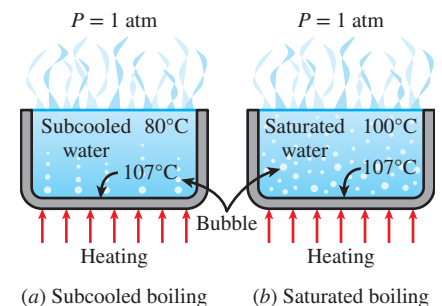


FIGURE 10–4

Classification of boiling on the basis of the presence of bulk liquid temperature.

10-2 ■ POOL BOILING

So far we presented some general discussions on boiling. Now we turn our attention to the physical mechanisms involved in *pool boiling*, that is, the boiling of stationary fluids. In pool boiling, the fluid is not forced to flow by a mover such as a pump, and any motion of the fluid is due to natural convection currents and the motion of the bubbles under the influence of buoyancy.

As a familiar example of pool boiling, consider the boiling of tap water in a pan on top of a stove. The water is initially at about 15°C, far below the saturation temperature of 100°C at standard atmospheric pressure. At the early stages of boiling, you will not notice anything significant except some bubbles that stick to the surface of the pan. These bubbles are caused by the release of air molecules dissolved in liquid water and should not be confused with vapor bubbles. As the water temperature rises, you will notice chunks of liquid water rolling up and down as a result of natural convection currents, followed by the first vapor bubbles forming at the bottom surface of the pan. These bubbles get smaller as they detach from the surface and start rising, and eventually collapse in the cooler water above. This is *subcooled boiling* since the bulk of the liquid water has not reached saturation temperature yet. The intensity of bubble formation increases as the water temperature rises further, and you will notice waves of vapor bubbles coming from the bottom and rising to the top when the water temperature reaches the saturation temperature (100°C at standard atmospheric conditions). This full-scale boiling is the *saturated boiling*.

Boiling Regimes and the Boiling Curve

Boiling is probably the most familiar form of heat transfer, yet it remains the least understood form. After hundreds of papers written on the subject, we still do not fully understand the process of bubble formation, and we must still rely on empirical or semi-empirical relations to predict the rate of boiling heat transfer.

The pioneering work on boiling was done in 1934 by S. Nukiyama, who used electrically heated nichrome and platinum wires immersed in liquids in his experiments. Nukiyama noticed that boiling takes different forms, depending on the value of the excess temperature ΔT_{excess} . Four different boiling regimes are observed: *natural convection boiling*, *nucleate boiling*, *transition boiling*, and *film boiling* (Fig. 10-5). These regimes are illustrated on the **boiling curve** in Fig. 10-6, which is a plot of boiling heat flux versus the excess temperature. Although the boiling curve given in this figure is for water, the general shape of the curve depends on the fluid–heating surface material combination and the fluid pressure, but it is practically independent of the geometry of the heating surface. We now describe each boiling regime in detail.

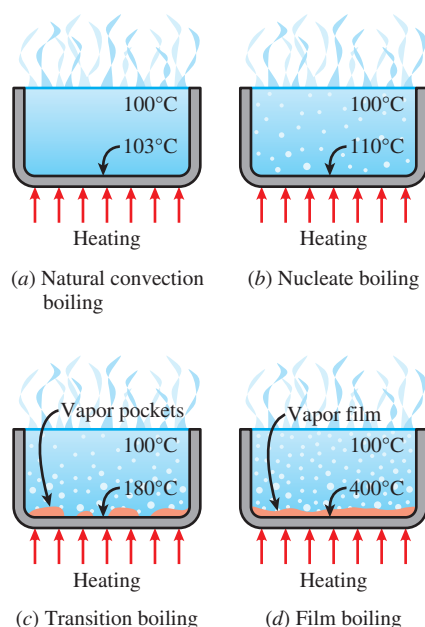


FIGURE 10-5

Different boiling regimes in pool boiling.

Natural Convection Boiling (to Point A on the Boiling Curve)

We know from thermodynamics that a pure substance at a specified pressure starts boiling when it reaches the saturation temperature at that pressure. But in practice we do not see any bubbles forming on the heating surface until the liquid is heated a few degrees above the saturation temperature (about 2 to 6°C for water). Therefore, the liquid is slightly *superheated* in this case (a *metastable* condition) and evaporates when it rises to the free surface. The fluid motion in

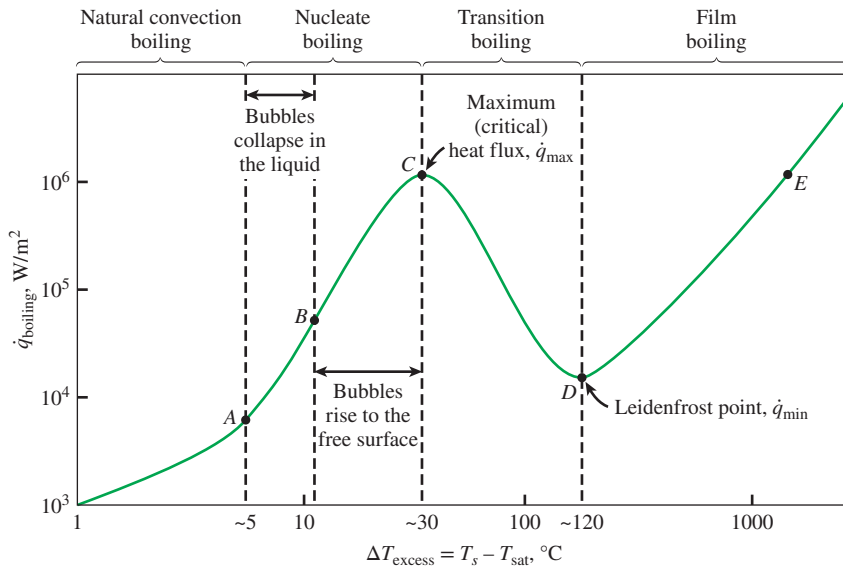


FIGURE 10–6
Typical boiling curve for water at
1 atm pressure.

this mode of boiling is governed by natural convection currents, and heat transfer from the heating surface to the fluid is by natural convection. For the conditions of Fig. 10–6, natural convection boiling ends at an excess temperature of about 5°C.

Nucleate Boiling (Between Points A and C)

The first bubbles start forming at point A of the boiling curve at various preferential sites on the heating surface. Point A is referred to as the **onset of nucleate boiling (ONB)**. The bubbles form at an increasing rate at an increasing number of nucleation sites as we move along the boiling curve toward point C. From Fig. 10–6, nucleate boiling exists in the range from about 5°C to about 30°C.

The nucleate boiling regime can be separated into two distinct regions. In region A–B ($5^\circ\text{C} \leq \Delta T_{\text{excess}} \leq 10^\circ\text{C}$), *isolated bubbles* are formed at various preferential nucleation sites on the heated surface. But these bubbles are dissipated in the liquid shortly after they separate from the surface. The space vacated by the rising bubbles is filled by the liquid in the vicinity of the heater surface, and the process is repeated. The stirring and agitation caused by the entrainment of the liquid to the heater surface is primarily responsible for the increased heat transfer coefficient and heat flux in this region of nucleate boiling.

In region B–C ($10^\circ\text{C} \leq \Delta T_{\text{excess}} \leq 30^\circ\text{C}$), the heater temperature is further increased, and bubbles form at such great rates at such a large number of nucleation sites that they form numerous *continuous columns of vapor* in the liquid. These bubbles move all the way up to the free surface, where they break up and release their vapor content. The large heat fluxes obtainable in this region are caused by the combined effect of liquid entrainment and evaporation.

At large values of ΔT_{excess} , the rate of evaporation at the heater surface reaches such high values that a large fraction of the heater surface is covered by bubbles, making it difficult for the liquid to reach the heater surface and wet it. Consequently, the heat flux increases at a lower rate with increasing ΔT_{excess} , and it reaches a maximum at point C. The heat flux at this point is called the **critical** (or **maximum**) **heat flux (CHF)**, $\dot{q}_{\text{max}}$. For water, the critical heat flux exceeds 1 MW/m². From Newton's law of cooling, the heat



(a)



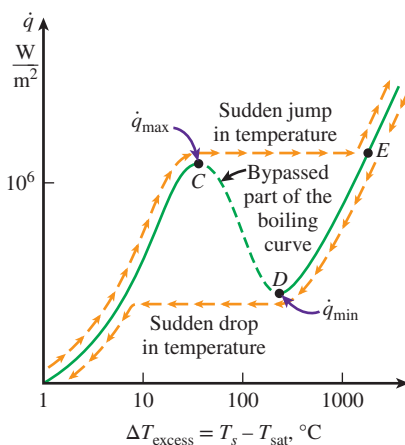
(b)

FIGURE 10–7

Nature of bubble formation and bubble motion associated with boiling.

(a) ©David Chasey/Getty Images

(b) ©McGraw-Hill Education/Stephen Frisch, photographer

**FIGURE 10–8**

The actual boiling curve obtained with heated platinum wire in water as the heat flux is increased and then decreased.

transfer coefficient at point C is $h = \dot{q}_{\text{max}} / \Delta T_{\text{excess}} = (10^6 \text{ W/m}^2) / 30 \text{ K} = 3.3 \times 10^4 \text{ W/m}^2 \cdot \text{K}$. For the entire nucleate boiling range of Fig. 10–6, the heat transfer coefficient ranges from about 2000 to 30,000 $\text{W/m}^2 \cdot \text{K}$, which is within the range of values tabulated in Table 1–5 for boiling.

Nucleate boiling is the most desirable boiling regime in practice because high heat transfer rates can be achieved in this regime with relatively small values of ΔT_{excess} , typically under 30°C for water. The photographs in Fig. 10–7 show the nature of bubble formation and bubble motion associated with boiling.

Transition Boiling (Between Points C and D)

As the heater temperature and thus the ΔT_{excess} is increased past point C, the heat flux decreases, as shown in Fig. 10–6. This is because a large fraction of the heater surface is covered by a vapor film, which acts as an insulation due to the low thermal conductivity of the vapor relative to that of the liquid. In the transition boiling regime, both nucleate and film boiling partially occur. Nucleate boiling at point C is completely replaced by film boiling at point D. Operation in the transition boiling regime, which is also called the *unstable film boiling regime*, is avoided in practice. For water, transition boiling occurs over the excess temperature range from about 30°C to about 120°C .

Film Boiling (Beyond Point D)

In this region, the heater surface is completely covered by a continuous stable vapor film. Point D, where the heat flux reaches a minimum, is called the **Leidenfrost point**, in honor of J. C. Leidenfrost, who observed in 1756 that liquid droplets on a very hot surface jump around and slowly boil away. The presence of a vapor film between the heater surface and the liquid is responsible for the low heat transfer rates in the film boiling region. The heat transfer rate increases with increasing excess temperature as a result of heat transfer from the heated surface to the liquid through the vapor film by radiation, which becomes significant at high temperatures.

A typical boiling process does not follow the boiling curve beyond point C, as Nukiyama has observed during his experiments. Nukiyama noticed, with surprise, that when the power applied to the nichrome wire immersed in water exceeded $\dot{q}_{\text{max}}$ even slightly, the wire temperature increased suddenly to the melting point of the wire (1500 K), and *burnout* occurred beyond his control. When he repeated the experiments with platinum wire, which has a much higher melting point (2045 K), he was able to avoid burnout and maintain heat fluxes higher than $\dot{q}_{\text{max}}$. When he gradually reduced power, he obtained the cooling curve shown in Fig. 10–8 with a sudden drop in excess temperature when $\dot{q}_{\text{min}}$ is reached. Note that the boiling process cannot follow the transition boiling part of the boiling curve past point C unless the power applied is reduced suddenly.

The *burnout phenomenon* in boiling can be explained as follows: In order to move beyond point C where $\dot{q}_{\text{max}}$ occurs, we must increase the heater surface temperature T_s . To increase T_s , however, we must increase the heat flux. But the fluid cannot receive this increased energy at an excess temperature just beyond point C. Therefore, the heater surface ends up absorbing the increased energy, causing the heater surface temperature T_s to rise. But the fluid can receive even less energy at this increased excess temperature, causing the heater surface temperature T_s to rise even further. This continues until the surface temperature reaches a point at which it no longer rises and the heat supplied can be transferred

to the fluid steadily. This is point *E* on the boiling curve, which corresponds to very high surface temperatures. Therefore, any attempt to increase the heat flux beyond $\dot{q}_{\max}$ will cause the operation point on the boiling curve to jump suddenly from point *C* to point *E*. However, surface temperature that corresponds to point *E* is beyond the melting point of most heater materials, and *burnout* occurs. Therefore, point *C* on the boiling curve is also called the **burnout point**, and the heat flux at this point is the **burnout heat flux** (Fig. 10–9).

Most boiling heat transfer equipment in practice operates slightly below $\dot{q}_{\max}$ to avoid any disastrous burnout. However, in cryogenic applications involving fluids with very low boiling points such as oxygen and nitrogen, point *E* usually falls below the melting point of the heater materials, and steady film boiling can be used in those cases without any danger of burnout.

Heat Transfer Correlations in Pool Boiling

Boiling regimes discussed above differ considerably in their character, and thus different heat transfer relations need to be used for different boiling regimes (Fig. 10–10). In the *natural convection boiling* regime ($\Delta T_{\text{excess}} \leq 5^\circ\text{C}$), boiling is governed by natural convection currents, and heat transfer rates in this case can be determined accurately using natural convection relations presented in Chap. 9.

Nucleate Boiling

In the *nucleate boiling* regime ($5^\circ\text{C} \leq \Delta T_{\text{excess}} \leq 30^\circ\text{C}$), the rate of heat transfer strongly depends on the nature of nucleation (the number of active nucleation sites on the surface, the rate of bubble formation at each site, etc.), which is difficult to predict. The type and the condition of the heated surface also affect the heat transfer. These complications made it difficult to develop theoretical relations for heat transfer in the nucleate boiling regime, and we had to rely on relations based on experimental data. The most widely used correlation for the rate of heat transfer in the nucleate boiling regime was proposed in 1952 by Rohsenow and expressed as

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3 \quad (10-2)$$

where

$\dot{q}_{\text{nucleate}}$ = nucleate boiling heat flux, W/m^2

μ_l = viscosity of the liquid, $\text{kg/m}\cdot\text{s}$

h_{fg} = enthalpy of vaporization, J/kg

g = gravitational acceleration, m/s^2

ρ_l = density of the liquid, kg/m^3

ρ_v = density of the vapor, kg/m^3

σ = surface tension of liquid–vapor interface, N/m

c_{pl} = specific heat of the liquid, $\text{J/kg}\cdot^\circ\text{C}$

T_s = surface temperature of the heater, $^\circ\text{C}$

T_{sat} = saturation temperature of the fluid, $^\circ\text{C}$

C_{sf} = experimental constant that depends on surface–fluid combination

Pr_l = Prandtl number of the liquid

n = experimental constant that depends on the fluid

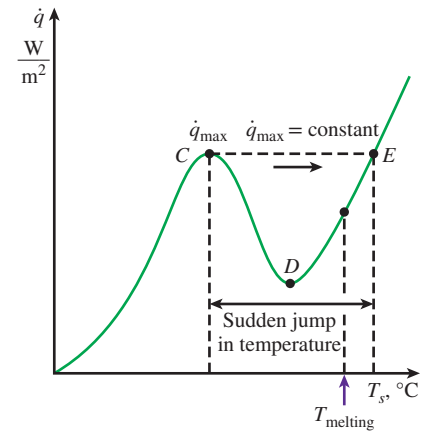


FIGURE 10–9

An attempt to increase the boiling heat flux beyond the critical value often causes the temperature of the heating element to jump suddenly to a value that is above the melting point, resulting in *burnout*.

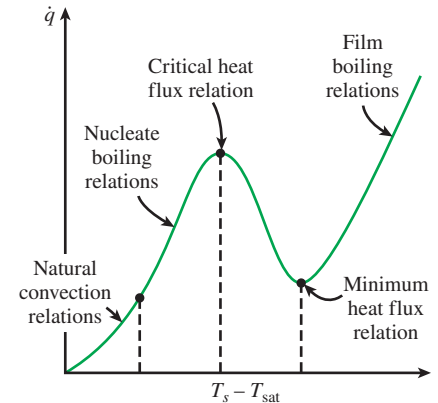


FIGURE 10–10

Different relations are used to determine the heat flux in different boiling regimes.

$$\begin{aligned}
 \dot{q} &= \left(\frac{\text{kg}}{\text{m} \cdot \text{s}} \right) \left(\frac{\text{J}}{\text{kg}} \right) \\
 &\times \left(\frac{\frac{\text{m}}{\text{s}^2} \frac{\text{kg}}{\text{m}^3}}{\frac{\text{N}}{\text{m}}} \right)^{1/2} \left(\frac{\frac{\text{J}}{\text{kg} \cdot ^\circ\text{C}}}{\frac{\text{J}}{\text{kg}}} \right)^3 \\
 &= \frac{\text{W}}{\text{m}} \left(\frac{1}{\text{m}^2} \right)^{1/2} (1)^3 \\
 &= \text{W/m}^2
 \end{aligned}$$

FIGURE 10–11

Equation 10–2 gives the boiling heat flux in W/m^2 when the quantities are expressed in the units specified in their descriptions.

TABLE 10–1

Surface tension of liquid–vapor interface for water

$T, ^\circ\text{C}$	$\sigma, \text{N/m}^*$
0	0.0757
20	0.0727
40	0.0696
60	0.0662
80	0.0627
100	0.0589
120	0.0550
140	0.0509
160	0.0466
180	0.0422
200	0.0377
220	0.0331
240	0.0284
260	0.0237
280	0.0190
300	0.0144
320	0.0099
340	0.0056
360	0.0019
374	0.0

*Multiply by 0.06852 to convert to lbf/ft or by 2.2046 to convert to lbfm/s^2 .

It can be shown easily that using property values in the specified units in the Rohsenow equation produces the desired unit W/m^2 for the boiling heat flux, thus saving one from having to go through tedious unit manipulations (Fig. 10–11).

The surface tension at the vapor–liquid interface is given in Table 10–1 for water and Table 10–2 for some other fluids. Experimentally determined values of the constant C_{sf} and n (fluid-dependent experimental constant) are given in Table 10–3 for various fluid–surface combinations. These values can be used for *any geometry* since it is found that the rate of heat transfer during nucleate boiling is essentially independent of the geometry and orientation of the heated surface. The fluid properties in Eq. 10–2 are to be evaluated at the saturation temperature T_{sat} .

The *condition* of the heater surface greatly affects heat transfer, and the Rohsenow equation given above is applicable to *clean* and relatively *smooth* surfaces. The results obtained using the Rohsenow equation can be in error by $\pm 100\%$ for the heat transfer rate for a given excess temperature and by $\pm 30\%$ for the excess temperature for a given heat transfer rate. Therefore, care should be exercised in the interpretation of the results.

Recall from thermodynamics that the enthalpy of vaporization h_{fg} of a pure substance decreases with increasing pressure (or temperature) and reaches zero at the critical point. Noting that h_{fg} appears in the denominator of the Rohsenow equation, we should see a significant rise in the rate of heat transfer at *high pressures* during nucleate boiling. Values of h_{fg} for different substances are listed in Tables A–2 and A–9 to A–12.

Peak Heat Flux

In the design of boiling heat transfer equipment, it is extremely important for the designer to have a knowledge of the maximum heat flux in order to avoid the danger of burnout. The *maximum* (or *critical*) *heat flux* in nucleate pool boiling was determined theoretically by S. S. Kutateladze in Russia in 1948 and N. Zuber in the United States in 1958 using quite different approaches, and it is expressed as

$$\dot{q}_{\text{max}} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \quad (10-3)$$

where C_{cr} is a constant whose value depends on the heater geometry. Exhaustive experimental studies by Lienhard and his coworkers indicated that the value of C_{cr} is about 0.15. Specific values of C_{cr} for different heater geometries are listed in Table 10–4. Note that the heaters are classified as being large or small based on the value of the parameter L^* .

Equation 10–3 will give the maximum heat flux in W/m^2 if the properties are used in the units specified earlier in their descriptions following Eq. 10–2. The maximum heat flux is independent of the fluid–heating surface combination, as well as the viscosity, thermal conductivity, and the specific heat of the liquid.

Note that ρ_v increases but σ and h_{fg} decrease with increasing pressure, and thus the change in $\dot{q}_{\text{max}}$ with pressure depends on which effect dominates. The experimental studies of Cichelli and Bonilla indicate that $\dot{q}_{\text{max}}$ increases with pressure up to about one-third of the critical pressure and then starts to decrease and becomes zero at the critical pressure. Also note that $\dot{q}_{\text{max}}$ is proportional to h_{fg} , and large maximum heat fluxes can be obtained using fluids with a large enthalpy of vaporization, such as water.

TABLE 10-2

Surface tension of some fluids

Substance and Temp. Range	Surface Tension, σ , N/m* (T in °C)
Ammonia, -75 to -40°C :	$0.0264 + 0.000223T$
Benzene, 10 to 80°C :	$0.0315 - 0.000129T$
Butane, -70 to -20°C :	$0.0149 - 0.000121T$
Carbon dioxide, -30 to -20°C :	$0.0043 - 0.000160T$
Ethyl alcohol, 10 to 70°C :	$0.0241 - 0.000083T$
Mercury, 5 to 200°C :	$0.4906 - 0.000205T$
Methyl alcohol, 10 to 60°C :	$0.0240 - 0.000077T$
Pentane, 10 to 30°C :	$0.0183 - 0.000110T$
Propane, -90 to -10°C :	$0.0092 - 0.000087T$

Source: From Suryanarayana, 1995, originally based on data from Jasper, 1972.

*Multiply by 0.06852 to convert to lbf/ft or by 2.2046 to convert to lbf/m².**TABLE 10-3**Values of the coefficient C_{sf} and n for various fluid–surface combinations

Fluid–Heating Surface Combination	C_{sf}	n
Water–copper (polished)	0.0130	1.0
Water–copper (scored)	0.0068	1.0
Water–stainless steel (mechanically polished)	0.0130	1.0
Water–stainless steel (ground and polished)	0.0060	1.0
Water–stainless steel (Teflon pitted)	0.0058	1.0
Water–stainless steel (chemically etched)	0.0130	1.0
Water–brass	0.0060	1.0
Water–nickel	0.0060	1.0
Water–platinum	0.0130	1.0
n -Pentane–copper (polished)	0.0154	1.7
n -Pentane–chromium	0.0150	1.7
Benzene–chromium	0.1010	1.7
Ethyl alcohol–chromium	0.0027	1.7
Carbon tetrachloride–copper	0.0130	1.7
Isopropanol–copper	0.0025	1.7

TABLE 10-4Values of the coefficient C_{cr} for use in Eq. 10-3 for maximum heat flux (dimensionless parameter $L^* = L[g(\rho_l - \rho_v)/\sigma]^{1/2}$)

Heater Geometry	C_{cr}	Charac. Dimension of Heater, L	Range of L^*
Large horizontal flat heater	0.149	Width or diameter	$L^* > 27$
Small horizontal flat heater ¹	$18.9K_1$	Width or diameter	$9 < L^* < 20$
Large horizontal cylinder	0.12	Radius	$L^* > 1.2$
Small horizontal cylinder	$0.12L^{*-0.25}$	Radius	$0.15 < L^* < 1.2$
Large sphere	0.11	Radius	$L^* > 4.26$
Small sphere	$0.227L^{*-0.5}$	Radius	$0.15 < L^* < 4.26$

¹ $K_1 = \sigma[g(\rho_l - \rho_v)A_{\text{heater}}]$

Minimum Heat Flux

Minimum heat flux, which occurs at the Leidenfrost point, is of practical interest since it represents the lower limit for the heat flux in the film boiling regime. Using the stability theory, Zuber (1958) derived the following expression for the minimum heat flux for a *large horizontal plate*,

$$\dot{q}_{\min} = 0.09\rho_v h_{fg} \left[\frac{\sigma g(\rho_l - \rho_v)}{(\rho_l + \rho_v)^2} \right]^{1/4} \quad (10-4)$$

where the constant 0.09 was determined by Berensen in 1961. He replaced the theoretically determined value of $\frac{\pi}{24}$ with 0.09 to match the experimental data better. Still, this relation can be in error by 50 percent or more.

Operation in the *transition boiling* regime ($30^\circ\text{C} \leq \Delta T_{\text{excess}} \leq 120^\circ\text{C}$) is normally avoided in the design of heat transfer equipment, and thus no major attempt has been made to develop general correlations for boiling heat transfer in this regime. However, the upper (*peak heat flux*, $\dot{q}_{\max}$) and the lower (*minimum heat flux*, $\dot{q}_{\min}$) limits of this region are of interest to heat transfer equipment designers.

Film Boiling

Using an analysis similar to Nusselt's theory on filmwise condensation presented in the next section, Bromley (1950) developed a theory for the prediction of heat flux for stable film boiling on the outside of a horizontal cylinder. The heat flux for film boiling on a *horizontal cylinder or sphere* of diameter D is given by

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \quad (10-5)$$

where k_v is the thermal conductivity of the vapor in $\text{W/m}\cdot\text{K}$ and

$$C_{\text{film}} = \begin{cases} 0.62 & \text{for horizontal cylinders} \\ 0.67 & \text{for spheres} \end{cases}$$

Other properties are as listed before in connection with Eq. 10-2. We used a modified latent heat of vaporization in Eq. 10-5 to account for the heat transfer associated with the superheating of the vapor.

The *vapor* properties are to be evaluated at the *film temperature* $T_f = (T_s + T_{\text{sat}})/2$, which is the *average temperature* of the vapor film. The liquid properties and h_{fg} are to be evaluated at the saturation temperature at the specified pressure. Again, this relation gives the film boiling heat flux in W/m^2 if the properties are used *in the units specified* earlier in their descriptions following Eq. 10-2.

At high surface temperatures (typically above 300°C), heat transfer across the vapor film by *radiation* becomes significant and needs to be considered (Fig. 10-12). Treating the vapor film as a transparent medium sandwiched between two large parallel plates and approximating the liquid as a blackbody, *radiation heat transfer* can be determined from

$$\dot{q}_{\text{rad}} = \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \quad (10-6)$$

where ε is the emissivity of the heating surface and $\sigma = 5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4$ is the Stefan–Boltzmann constant. Note that the temperature in this case must be expressed in K, not $^\circ\text{C}$, and that surface tension and the Stefan–Boltzmann constant share the same symbol.

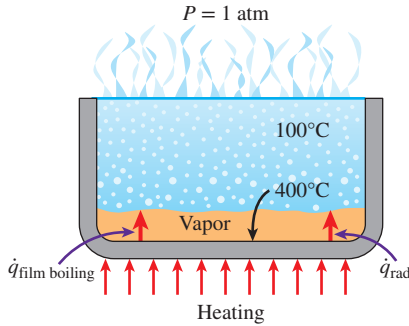


FIGURE 10-12

At high heater surface temperatures, radiation heat transfer becomes significant during film boiling.

You may be tempted to simply add the convection and radiation heat transfers to determine the total heat transfer during film boiling. However, these two mechanisms of heat transfer adversely affect each other, causing the total heat transfer to be less than their sum. For example, the radiation heat transfer from the surface to the liquid enhances the rate of evaporation, and thus the thickness of the vapor film, which impedes convection heat transfer. For $\dot{q}_{\text{rad}} < \dot{q}_{\text{film}}$, Bromley (1950) determined that the relation

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4}\dot{q}_{\text{rad}} \quad (10-7)$$

correlates experimental data well.

Note that the gravitational acceleration g , whose value is approximately 9.81 m/s^2 at sea level, appears in all of the preceding relations for boiling heat transfer. The effects of low and high gravity (as encountered in aerospace applications and turbomachinery) are studied experimentally. The studies confirm that the critical heat flux and heat flux in film boiling are proportional to $g^{1/4}$. However, they indicate that heat flux in nucleate boiling is practically independent of gravity g , instead of being proportional to $g^{1/2}$, as dictated by Eq. 10-2.

Enhancement of Heat Transfer in Pool Boiling

The pool boiling heat transfer relations given to this point apply to smooth surfaces. Next we discuss some methods to enhance heat transfer in pool boiling.

We pointed out earlier that the rate of heat transfer in the nucleate boiling regime strongly depends on the number of active nucleation sites on the surface and the rate of bubble formation at each site. Therefore, any modification that enhances *nucleation* on the heating surface will also enhance *heat transfer* in nucleate boiling. It is observed that *irregularities* on the heating surface, including roughness and dirt, serve as additional nucleation sites during boiling, as shown in Fig. 10-13. For example, the first bubbles in a pan filled with water are most likely to form at the *scratches* at the bottom surface. These scratches act like “nests” for the bubbles to form and thus increase the rate of bubble formation. Berensen has shown that heat flux in the nucleate boiling regime can be increased by a factor of 10 by *roughening* the heating surface. However, these high heat transfer rates cannot be sustained for long since the effect of surface roughness is observed to decay with time and the heat flux to drop eventually to values encountered on smooth surfaces. The effect of surface roughness is negligible on the maximum (critical) heat flux and the heat flux in film boiling.

Surfaces that provide enhanced heat transfer in nucleate boiling *permanently* are being manufactured and are available in the market. Enhancement in nucleation and thus heat transfer in such special surfaces is achieved either by *coating* the surface with a thin layer (much less than 1 mm) of very porous material or by *forming cavities* on the surface mechanically to facilitate continuous vapor formation. Such surfaces are reported to enhance heat transfer in the nucleate boiling regime by a factor of up to 10 and the maximum heat flux by a factor of 3. The enhancement provided by one such material prepared by machine roughening, the thermoexcel-E, is shown in Fig. 10-14. The use of finned surfaces is also known to enhance nucleate boiling heat transfer and the maximum heat flux.

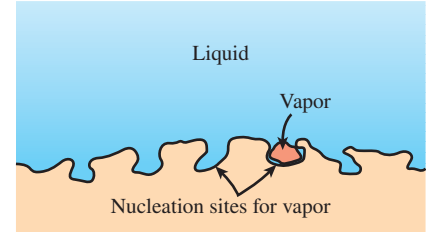


FIGURE 10-13

The cavities on a rough surface act as nucleation sites and enhance boiling heat transfer.

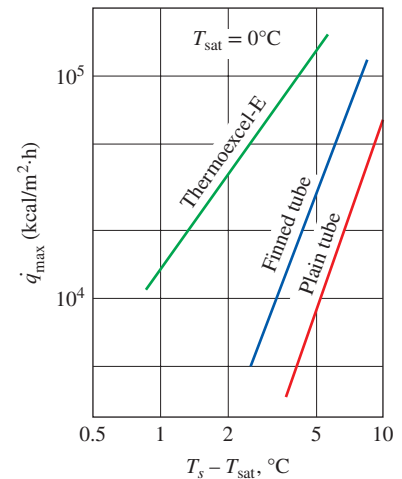
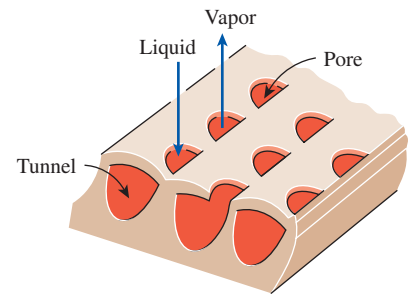


FIGURE 10-14

The enhancement of boiling heat transfer in Freon-12 by a mechanically roughened surface, thermoexcel-E.

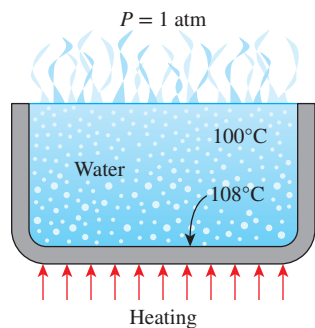


FIGURE 10–15

Schematic for Example 10–1.

Boiling heat transfer can also be enhanced by other techniques such as *mechanical agitation* and *surface vibration*. These techniques are not practical, however, because of the complications involved.

EXAMPLE 10–1 Nucleate Boiling of Water in a Pan

Water is to be boiled at atmospheric pressure in a mechanically polished stainless steel pan placed on top of a heating unit, as shown in Fig. 10–15. The inner surface of the bottom of the pan is maintained at 108°C. If the diameter of the bottom of the pan is 30 cm, determine (a) the rate of heat transfer to the water and (b) the rate of evaporation of water.

SOLUTION Water is boiled at 1 atm pressure on a stainless steel surface. The rate of heat transfer to the water and the rate of evaporation of water are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the heater and the pan are negligible.

Properties The properties of water at the saturation temperature of 100°C are $\sigma = 0.0589$ N/m (Table 10–1) and, from Table A–9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.6 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg}\cdot\text{K}\end{aligned}$$

Also, $C_{sf} = 0.0130$ and $n = 1.0$ for the boiling of water on a mechanically polished stainless steel surface (Table 10–3). Note that we expressed the properties in units specified under Eq. 10–2 in connection with their definitions in order to avoid unit manipulations.

Analysis (a) The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 108 - 100 = 8^\circ\text{C}$, which is relatively low (less than 30°C). Therefore, nucleate boiling will occur. The heat flux in this case can be determined from the Rohsenow relation to be

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3 \\ &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81 \times (957.9 - 0.6)}{0.0589} \right]^{1/2} \\ &\quad \times \left(\frac{4217(108 - 100)}{0.0130(2257 \times 10^3) 1.75} \right)^3 \\ &= 7.21 \times 10^4 \text{ W/m}^2\end{aligned}$$

The surface area of the bottom of the pan is

$$A = \pi D^2/4 = \pi(0.3 \text{ m})^2/4 = 0.07069 \text{ m}^2$$

Then the rate of heat transfer during nucleate boiling becomes

$$\dot{Q}_{\text{boiling}} = A \dot{q}_{\text{nucleate}} = (0.07069 \text{ m}^2)(7.21 \times 10^4 \text{ W/m}^2) = \mathbf{5097 \text{ W}}$$

(b) The rate of evaporation of water is determined from

$$\dot{m}_{\text{evaporation}} = \frac{\dot{Q}_{\text{boiling}}}{h_{fg}} = \frac{5097 \text{ J/s}}{2257 \times 10^3 \text{ J/kg}} = \mathbf{2.26 \times 10^{-3} \text{ kg/s}}$$

That is, water in the pan will boil at a rate of more than 2 grams per second.

EXAMPLE 10–2 Peak Heat Flux in Nucleate Boiling

Water in a tank is to be boiled at sea level by a 1-cm-diameter nickel-plated steel heating element equipped with electrical resistance wires inside, as shown in Fig. 10–16. Determine the maximum heat flux that can be attained in the nucleate boiling regime and the surface temperature of the heater in that case.

SOLUTION Water is boiled at 1 atm pressure on a nickel-plated steel surface. The maximum heat flux and the surface temperature are to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $\sigma = 0.0589$ N/m (Table 10–1) and, from Table A–9,

$$\begin{aligned}\rho_l &= 957.9 \text{ kg/m}^3 & h_{fg} &= 2257 \times 10^3 \text{ J/kg} \\ \rho_v &= 0.6 \text{ kg/m}^3 & \mu_l &= 0.282 \times 10^{-3} \text{ kg/m}\cdot\text{s} \\ \text{Pr}_l &= 1.75 & c_{pl} &= 4217 \text{ J/kg}\cdot\text{K}\end{aligned}$$

Also, $C_{sf} = 0.0060$ and $n = 1.0$ for the boiling of water on a nickel-plated surface (Table 10–3). Note that we expressed the properties in units specified under Eqs. 10–2 and 10–3 in connection with their definitions in order to avoid unit manipulations.

Analysis The heating element in this case can be considered to be a short cylinder whose characteristic dimension is its radius. That is, $L = r = 0.005$ m. The dimensionless parameter L^* and the constant C_{cr} are determined from Table 10–4 to be

$$L^* = L \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} = (0.005) \left[\frac{(9.81)(957.9 - 0.6)}{0.0589} \right]^{1/2} = 2.00 > 1.2$$

which corresponds to $C_{cr} = 0.12$.

Then the maximum or critical heat flux is determined from Eq. 10–3 to be

$$\begin{aligned}\dot{q}_{\max} &= C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4} \\ &= 0.12 (2257 \times 10^3) [0.0589 \times 9.81 \times (0.6)^2 (957.9 - 0.6)]^{1/4} \\ &= \mathbf{1.017 \times 10^6 \text{ W/m}^2}\end{aligned}$$

The Rohsenow relation, which gives the nucleate boiling heat flux for a specified surface temperature, can also be used to determine the surface temperature when the heat flux is given. Substituting the maximum heat flux into Eq. 10–2 together with other properties gives

$$\begin{aligned}\dot{q}_{\text{nucleate}} &= \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3 \\ 1.017 \times 10^6 &= (0.282 \times 10^{-3})(2257 \times 10^3) \left[\frac{9.81(957.9 - 0.6)}{0.0589} \right]^{1/2} \\ &\quad \times \left[\frac{4217(T_s - 100)}{0.0130 (2257 \times 10^3) 1.75} \right]^3 \\ T_s &= \mathbf{119^\circ\text{C}}\end{aligned}$$

Discussion Note that heat fluxes on the order of 1 MW/m² can be obtained in nucleate boiling with a temperature difference of less than 20°C.

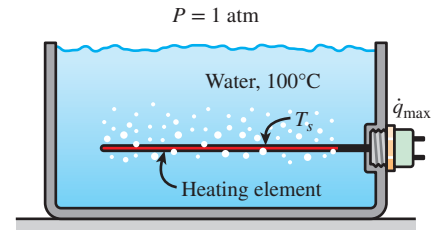


FIGURE 10–16
Schematic for Example 10–2.

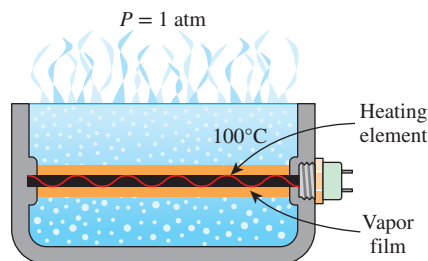


FIGURE 10-17

Schematic for Example 10-3.

EXAMPLE 10-3 Film Boiling of Water on a Heating Element

Water is boiled at atmospheric pressure by a horizontal polished copper heating element of diameter $D = 5$ mm and emissivity $\varepsilon = 0.05$ immersed in water, as shown in Fig. 10-17. If the surface temperature of the heating wire is 350°C , determine the rate of heat transfer from the wire to the water per unit length of the wire.

SOLUTION Water is boiled at 1 atm by a horizontal polished copper heating element. The rate of heat transfer to the water per unit length of the heater is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Heat losses from the boiler are negligible.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3$ J/kg and $\rho_l = 957.9$ kg/m³ (Table A-9). The properties of vapor at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 350)/2 = 225^\circ\text{C}$ are, from Table A-16,

$$\begin{aligned}\rho_v &= 0.444 \text{ kg/m}^3 & c_{pv} &= 1951 \text{ J/kg}\cdot\text{K} \\ \mu_v &= 1.75 \times 10^{-5} \text{ kg/m}\cdot\text{s} & k_v &= 0.0358 \text{ W/m}\cdot\text{K}\end{aligned}$$

Note that we expressed the properties in units that cancel each other in boiling heat transfer relations. Also note that we used vapor properties at 1 atm pressure from Table A-16 instead of the properties of saturated vapor from Table A-9 at 225°C since the latter are at the saturation pressure of 2.55 MPa.

Analysis The excess temperature in this case is $\Delta T = T_s - T_{\text{sat}} = 305 - 100 = 250^\circ\text{C}$, which is much larger than 30°C for water. Therefore, film boiling will occur. The film boiling heat flux in this case can be determined from Eq. 10-5 to be

$$\begin{aligned}\dot{q}_{\text{film}} &= 0.62 \left[\frac{gk_v^3 \rho_v (\rho_l - \rho_v) [h_{fg} + 0.4c_{pv}(T_s - T_{\text{sat}})]}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} (T_s - T_{\text{sat}}) \\ &= 0.62 \left[\frac{9.81(0.0358)^3 (0.444)(957.9 - 0.441) \times [(2257 \times 10^3) + 0.4 \times 1951(250)]}{(1.75 \times 10^{-5})(5 \times 10^{-3})(250)} \right]^{1/4} \times 250 \\ &= 5.93 \times 10^4 \text{ W/m}^2\end{aligned}$$

The radiation heat flux is determined from Eq. 10-6 to be

$$\begin{aligned}\dot{q}_{\text{rad}} &= \varepsilon \sigma (T_s^4 - T_{\text{sat}}^4) \\ &= (0.05)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(350 + 273 \text{ K})^4 - (100 + 273 \text{ K})^4] \\ &= 372 \text{ W/m}^2\end{aligned}$$

Note that heat transfer by radiation is negligible in this case because of the low emissivity of the surface and the relatively low surface temperature of the heating element. Then the total heat flux becomes (Eq. 10-7)

$$\dot{q}_{\text{total}} = \dot{q}_{\text{film}} + \frac{3}{4} \dot{q}_{\text{rad}} = 5.93 \times 10^4 + \frac{3}{4} \times 372 = 5.96 \times 10^4 \text{ W/m}^2$$

Finally, the rate of heat transfer from the heating element to the water is determined by multiplying the heat flux by the heat transfer surface area,

$$\begin{aligned}\dot{Q}_{\text{total}} &= A\dot{q}_{\text{total}} = (\pi DL)\dot{q}_{\text{total}} \\ &= (\pi \times 0.005 \text{ m} \times 1 \text{ m})(5.96 \times 10^4 \text{ W/m}^2) \\ &= \mathbf{936 \text{ W}}\end{aligned}$$

Discussion Note that the 5-mm-diameter copper heating element consumes about 1 kW of electric power per unit length in steady operation in the film boiling regime. This energy is transferred to the water through the vapor film that forms around the wire.

10-3 ■ FLOW BOILING

The pool boiling we considered so far involves a pool of seemingly motionless liquid, with vapor bubbles rising to the top as a result of buoyancy effects. In **flow boiling**, the fluid is forced to move by an external source such as a pump as it undergoes a phase-change process. The boiling in this case exhibits the combined effects of convection and pool boiling. The flow boiling is also classified as either *external flow boiling* or *internal flow boiling* depending on whether the fluid is forced to flow over a heated surface or inside a heated tube.

External flow boiling over a plate or cylinder is similar to pool boiling, but the added motion increases both the nucleate boiling heat flux and the maximum heat flux considerably, as shown in Fig. 10–18. Note that the higher the velocity, the higher the nucleate boiling heat flux and the maximum heat flux. In experiments with water, maximum heat flux values as high as 35 MW/m² have been obtained (compare this to the pool boiling value of 1.02 MW/m² at 1 atm pressure) by increasing the fluid velocity.

Internal flow boiling, commonly referred to as *two-phase flow*, is much more complicated in nature because there is no free surface for the vapor to escape, and thus both the liquid and the vapor are forced to flow together. The two-phase flow in a tube exhibits different flow boiling regimes, depending on the relative amounts of the liquid and the vapor phases. This complicates the analysis even further.

The different stages encountered in flow boiling in a heated tube are illustrated in Fig. 10–19 together with the variation of the heat transfer coefficient along the tube. Initially, the liquid is subcooled, and heat transfer to the liquid is by *forced convection*. Then bubbles start forming on the inner surfaces of the tube, and the detached bubbles are drafted into the mainstream. This gives the fluid flow a bubbly appearance, and thus the name *bubbly flow regime*. As the fluid is heated further, the bubbles grow in size and eventually coalesce into slugs of vapor. Up to half of the volume in the tube in this *slug-flow regime* is occupied by vapor. After a while the core of the flow consists of vapor only, and the liquid is confined only in the annular space between the vapor core and the tube walls. This is the *annular-flow regime*, and very high heat transfer coefficients are realized in this regime.

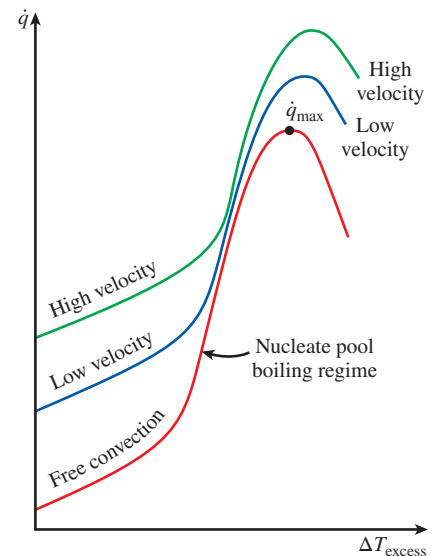
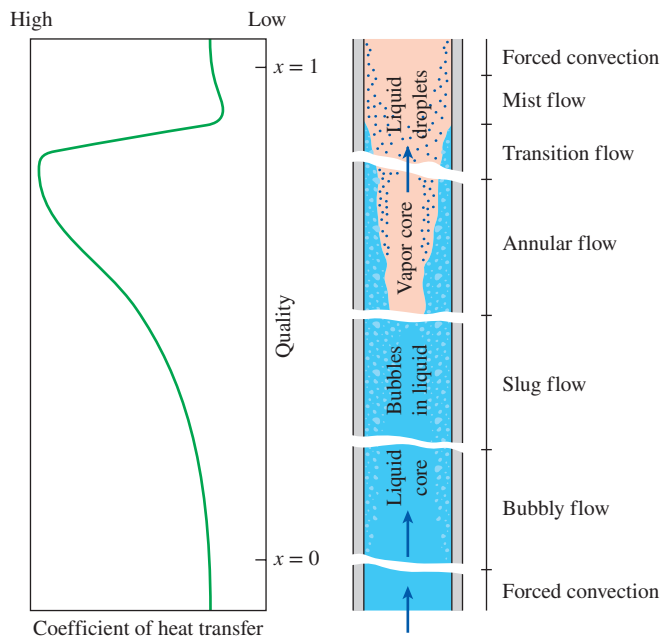


FIGURE 10-18

The effect of forced convection on external flow boiling for different flow velocities.

**FIGURE 10–19**

Different flow regimes encountered in flow boiling in a tube under forced convection.

As the heating continues, the annular liquid layer gets thinner and thinner, and eventually dry spots start to appear on the inner surfaces of the tube. The appearance of dry spots is accompanied by a sharp decrease in the heat transfer coefficient. This *transition regime* continues until the inner surface of the tube is completely dry. Any liquid at this moment is in the form of droplets suspended in the vapor core, which resembles a mist, and we have a *mist-flow regime* until all the liquid droplets are vaporized. At the end of the mist-flow regime we have saturated vapor, which becomes superheated with any further heat transfer.

Note that the tube contains a liquid before the bubbly flow regime and a vapor after the mist-flow regime. Heat transfer in those two cases can be determined using the appropriate relations for single-phase convection heat transfer. Many correlations are proposed for the determination of heat transfer in the two-phase flow (bubbly flow, slug-flow, annular-flow, and mist-flow) cases, but they are beyond the scope of this introductory text. A crude estimate for heat flux in flow boiling can be obtained by simply adding the forced convection and pool boiling heat fluxes.

10–4 ■ CONDENSATION HEAT TRANSFER

Condensation occurs when the temperature of a vapor is reduced *below* its saturation temperature T_{sat} . This is usually done by bringing the vapor into contact with a solid surface whose temperature T_s is below the saturation temperature T_{sat} of the vapor. But condensation can also occur on the free surface of a liquid or even in a gas when the temperature of the liquid or the gas to which the vapor is exposed is below T_{sat} . In the latter case, the liquid droplets suspended in the gas form a fog. In this chapter, we consider condensation on solid surfaces only.

Two distinct forms of condensation are observed: *film condensation* and *dropwise condensation*. In **film condensation**, the condensate wets the surface and forms a liquid film on the surface that slides down under the influence of gravity. The thickness of the liquid film increases in the flow direction as more vapor condenses on the film. This is how condensation normally occurs in practice. In **dropwise condensation**, the condensed vapor forms droplets on the surface instead of a continuous film, and the surface is covered by countless droplets of varying diameters (Fig. 10–20).

In film condensation, the surface is blanketed by a liquid film of increasing thickness, and this “liquid wall” between the solid surface and the vapor serves as a *resistance* to heat transfer. The heat of vaporization h_{fg} released as the vapor condenses must pass through this resistance before it can reach the solid surface and be transferred to the medium on the other side. In dropwise condensation, however, the droplets slide down when they reach a certain size, clearing the surface and exposing it to vapor. There is no liquid film in this case to resist heat transfer. As a result, heat transfer rates that are more than 10 times larger than those associated with film condensation can be achieved with dropwise condensation. Therefore, dropwise condensation is the preferred mode of condensation in heat transfer applications, and people have long tried to achieve sustained dropwise condensation by using various vapor additives and surface coatings. These attempts have not been very successful, however, since the dropwise condensation achieved did not last long and converted to film condensation after some time. Therefore, it is common practice to be conservative and assume film condensation in the design of heat transfer equipment. For better heat transfer, it is desirable to use short surfaces because of the lower thermal resistance.

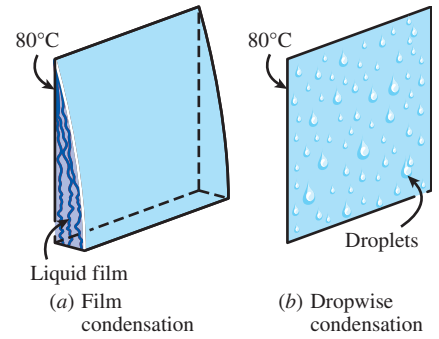


FIGURE 10–20

When a vapor is exposed to a surface at a temperature below T_{sat} , condensation in the form of a liquid film or individual droplets occurs on the surface.

10–5 ■ FILM CONDENSATION

We now consider film condensation on a vertical plate, as shown in Fig. 10–21. The liquid film starts forming at the top of the plate and flows downward under the influence of gravity. The thickness of the film δ *increases* in the flow direction x because of continued condensation at the liquid–vapor interface. Heat in the amount h_{fg} (the latent heat of vaporization) is released during condensation and is *transferred* through the film to the plate surface at temperature T_s . Note that T_s must be below the saturation temperature T_{sat} of the vapor for condensation to occur.

Typical velocity and temperature profiles of the condensate are also given in Fig. 10–21. Note that the *velocity* of the condensate at the wall is zero because of the “no-slip” condition and reaches a *maximum* at the liquid–vapor interface. The *temperature* of the condensate is T_{sat} at the interface and decreases gradually to T_s at the wall.

As was the case in forced convection involving a single phase, heat transfer in condensation also depends on whether the condensate flow is *laminar* or *turbulent*. Again the criterion for the flow regime is provided by the Reynolds number, which is defined as

$$\text{Re} = \frac{D_h \rho_l V_l}{\mu_l} = \frac{4A_c \rho_l V_l}{p \mu_l} = \frac{4\rho_l V_l \delta}{\mu_l} = \frac{4\dot{m}}{p \mu_l} \quad (10-8)$$

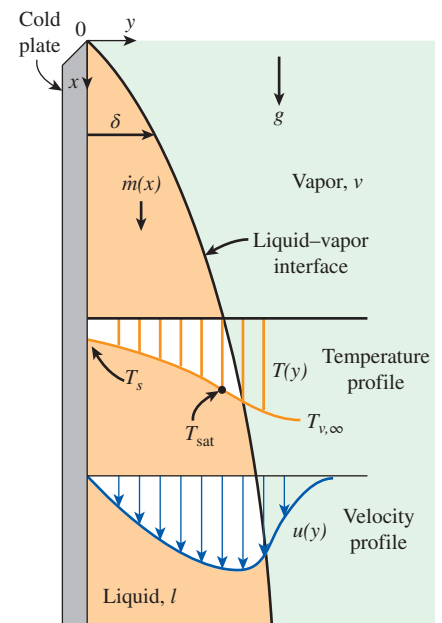


FIGURE 10–21

Film condensation on a vertical plate.

where

$D_h = 4A_c/p = 4\delta$ = hydraulic diameter of the condensate flow, m

p = wetted perimeter of the condensate, m

$A_c = p\delta$ = wetted perimeter $\times$ film thickness, m^2 , cross-sectional area of the condensate flow at the lowest part of the flow

ρ_l = density of the liquid, kg/m^3

μ_l = viscosity of the liquid, $\text{kg}/\text{m}\cdot\text{s}$

V_l = average velocity of the condensate at the lowest part of the flow, m/s

$\dot{m} = \rho_l V_l A_c$ = mass flow rate of the condensate at the lowest part, kg/s

The evaluation of the hydraulic diameter D_h for some common geometries is illustrated in Fig. 10–22. Note that the hydraulic diameter is again defined such that it reduces to the ordinary diameter for flow in a circular tube, as was done in Chap. 8 for internal flow, and it is equivalent to four times the thickness of the condensate film at the location where the hydraulic diameter is evaluated. That is, $D_h = 4\delta$.

The latent heat of vaporization h_{fg} is the heat released as a unit mass of vapor condenses, and it normally represents the heat transfer per unit mass of condensate formed during condensation. However, the condensate in an actual condensation process is cooled further to some average temperature between T_{sat} and T_s , releasing more heat in the process. Therefore, the actual heat transfer will be larger. Rohsenow showed in 1956 that the cooling of the liquid below the saturation temperature can be accounted for by replacing h_{fg} with the **modified latent heat of vaporization** h_{fg}^* , defined as

$$h_{fg}^* = h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) \quad (10-9a)$$

where c_{pl} is the specific heat of the liquid at the average film temperature.

We can have a similar argument for vapor that enters the condenser as **superheated vapor** at a temperature T_v instead of as saturated vapor. In this case, the vapor must be cooled first to T_{sat} before it can condense, and this heat must be transferred to the wall as well. The amount of heat released as a unit mass of superheated vapor at a temperature T_v is cooled to T_{sat} is simply

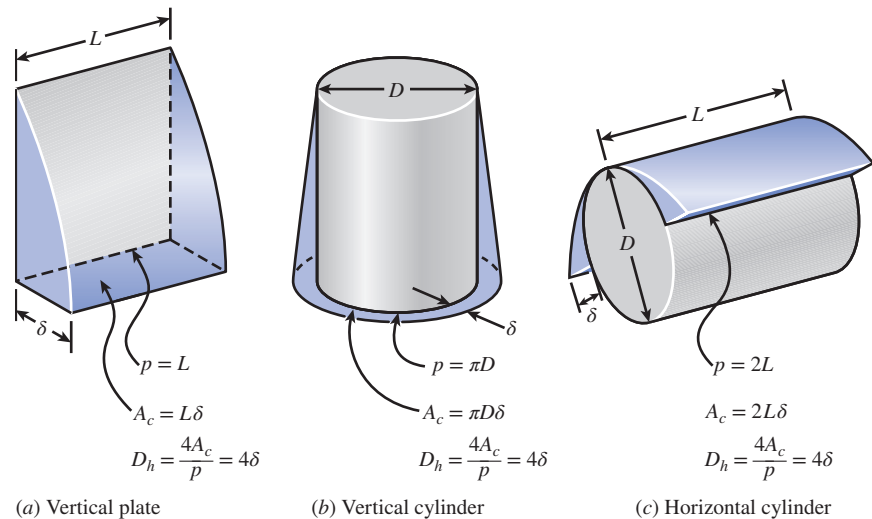


FIGURE 10–22

The wetted perimeter p , the condensate cross-sectional area A_c , and the hydraulic diameter D_h for some common geometries.

$c_{pv}(T_v - T_{\text{sat}})$, where c_{pv} is the specific heat of the vapor at the average temperature of $(T_v + T_{\text{sat}})/2$. The modified latent heat of vaporization in this case becomes

$$h_{fg}^* = h_{fg} + 0.68c_{pl}(T_{\text{sat}} - T_s) + c_{pv}(T_v - T_{\text{sat}}) \quad (10-9b)$$

With these considerations, the rate of heat transfer can be expressed as

$$\dot{Q}_{\text{conden}} = hA_s(T_{\text{sat}} - T_s) = \dot{m}h_{fg}^* \quad (10-10)$$

where A_s is the heat transfer area (the surface area on which condensation occurs). Solving for $\dot{m}$ from the equation above and substituting it into Eq. 10-8 gives yet another relation for the Reynolds number,

$$\text{Re} = \frac{4\dot{Q}_{\text{conden}}}{p\mu_l h_{fg}^*} = \frac{4A_s h(T_{\text{sat}} - T_s)}{p\mu_l h_{fg}^*} \quad (10-11)$$

This relation is convenient to use to determine the Reynolds number when the condensation heat transfer coefficient or the rate of heat transfer is known.

The temperature of the liquid film varies from T_{sat} on the liquid–vapor interface to T_s at the wall surface. Therefore, the properties of the liquid should be evaluated at the *film temperature* $T_f = (T_{\text{sat}} + T_s)/2$, which is approximately the *average* temperature of the liquid. The h_{fg}^* , however, should be evaluated at T_{sat} since it is not affected by the subcooling of the liquid.

Flow Regimes

The Reynolds number for condensation on the outer surfaces of vertical tubes or plates increases in the flow direction due to the increase of the liquid film thickness δ . The flow of liquid film exhibits *different regimes*, depending on the value of the Reynolds number. It is observed that the outer surface of the liquid film remains *smooth* and *wave-free* for about $\text{Re} \leq 30$, as shown in Fig. 10-23, and thus the flow is clearly *laminar*. Ripples or waves appear on the free surface of the condensate flow as the Reynolds number increases, and the condensate flow becomes fully *turbulent* at about $\text{Re} \approx 1800$. The condensate flow is called *wavy-laminar* in the range of $30 < \text{Re} < 1800$ and *turbulent* for $\text{Re} > 1800$. However, some disagreement exists about the value of Re at which the flow becomes wavy-laminar or turbulent.

Heat Transfer Correlations for Film Condensation

Next we discuss relations for the average heat transfer coefficient h for the case of *laminar* film condensation for various geometries.

1 Vertical Plates

Consider a vertical plate of height L and width b maintained at a constant temperature T_s that is exposed to vapor at the saturation temperature T_{sat} . The downward direction is taken as the positive x -direction with the origin placed at the top of the plate where condensation initiates, as shown in Fig. 10-24. The surface temperature is below the saturation temperature ($T_s < T_{\text{sat}}$) and thus the vapor condenses on the surface. The liquid film flows downward under the influence of gravity. The film thickness δ and thus the mass flow rate of the condensate increases with x as a result of continued condensation on the existing film. Then heat transfer from the vapor to the plate must occur through the film, which offers resistance to heat transfer. Obviously the thicker the film, the larger its thermal resistance and thus the lower the rate of heat transfer.

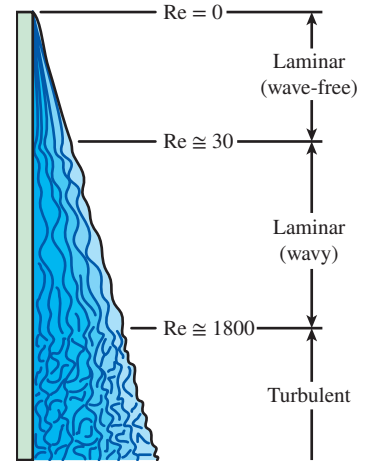


FIGURE 10-23

Flow regimes during film condensation on a vertical plate.

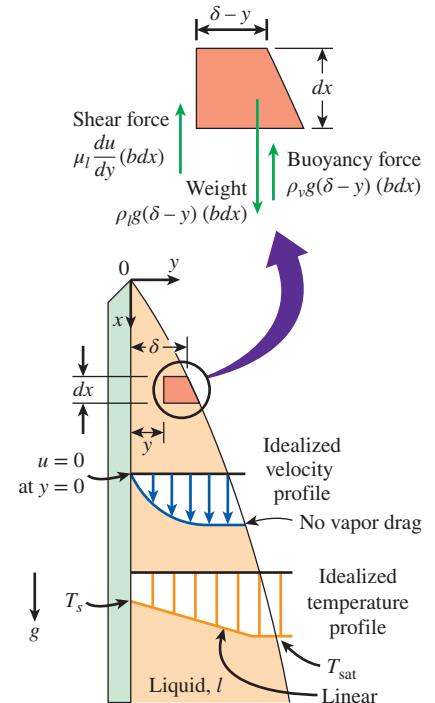


FIGURE 10-24

The volume element of condensate on a vertical plate considered in Nusselt's analysis.

The analytical relation for the heat transfer coefficient in film condensation on a vertical plate described above was first developed by Nusselt in 1916 under the following simplifying assumptions:

1. Both the plate and the vapor are maintained at *constant temperatures* of T_s and T_{sat} , respectively, and the temperature across the liquid film varies *linearly*.
2. Heat transfer across the liquid film is by pure *conduction* (no convection currents in the liquid film).
3. The velocity of the vapor is low (or zero) so that it exerts *no drag* on the condensate (no viscous shear on the liquid–vapor interface).
4. The flow of the condensate is *laminar* and the properties of the liquid are constant.
5. The acceleration of the condensate layer is negligible.

Then Newton's second law of motion for the volume element shown in Fig. 10–24 in the vertical x -direction can be written as

$$\sum F_x = ma_x = 0$$

since the acceleration of the fluid is zero. Noting that the only force acting downward is the weight of the liquid element, and the forces acting upward are the viscous shear (or fluid friction) force at the left and the buoyancy force, the force balance on the volume element becomes

$$F_{\text{downward} \downarrow} = F_{\text{upward} \uparrow}$$

Weight = Viscous shear force + Buoyancy force

$$\rho_l g(\delta - y)(b dx) = \mu_l \frac{du}{dy}(b dx) + \rho_v g(\delta - y)(b dx)$$

Canceling the plate width b and solving for du/dy gives

$$\frac{du}{dy} = \frac{g(\rho_l - \rho_v)(\delta - y)}{\mu_l}$$

Integrating from $y = 0$ where $u = 0$ (because of the no-slip boundary condition) to $y = y$ where $u = u(y)$ gives

$$u(y) = \frac{g(\rho_l - \rho_v)}{\mu_l} \left(y\delta - \frac{y^2}{2} \right) \quad (10-12)$$

The mass flow rate of the condensate at a location x , where the boundary layer thickness is δ , is determined from

$$\dot{m}(x) = \int_A \rho_l u(y) dA = \int_{y=0}^{\delta} \rho_l u(y) b dy \quad (10-13)$$

Substituting the $u(y)$ relation from Eq. 10–12 into Eq. 10–13 gives

$$\dot{m}(x) = \frac{gb\rho_l(\rho_l - \rho_v)\delta^3}{3\mu_l} \quad (10-14)$$

whose derivative with respect to x is

$$\frac{d\dot{m}}{dx} = \frac{gb\rho_l(\rho_l - \rho_v)\delta^2}{\mu_l} \frac{d\delta}{dx} \quad (10-15)$$

which represents the rate of condensation of vapor over a vertical distance dx . The rate of heat transfer from the vapor to the plate through the liquid film is simply equal to the heat released as the vapor is condensed and is expressed as

$$d\dot{Q} = h_{fg} dm = k_l (b dx) \frac{T_{\text{sat}} - T_s}{\delta} \rightarrow \frac{dm}{dx} = \frac{k_l b}{h_{fg}} \frac{T_{\text{sat}} - T_s}{\delta} \quad (10-16)$$

Equating Eqs. 10–15 and 10–16 for dm/dx to each other and separating the variables give

$$\delta^3 d\delta = \frac{\mu_l k_l (T_{\text{sat}} - T_s)}{g \rho_l (\rho_l - \rho_v) h_{fg}} dx \quad (10-17)$$

Integrating from $x = 0$ where $\delta = 0$ (the top of the plate) to $x = x$ where $\delta = \delta(x)$, the liquid film thickness at any location x is determined to be

$$\delta(x) = \left[\frac{4 \mu_l k_l (T_{\text{sat}} - T_s) x}{g \rho_l (\rho_l - \rho_v) h_{fg}} \right]^{1/4} \quad (10-18)$$

The heat transfer rate from the vapor to the plate at a location x can be expressed as

$$\dot{q}_x = h_x (T_{\text{sat}} - T_s) = k_l \frac{T_{\text{sat}} - T_s}{\delta} \rightarrow h_x = \frac{k_l}{\delta(x)} \quad (10-19)$$

Substituting the $\delta(x)$ expression from Eq. 10–18, the local heat transfer coefficient h_x is determined to be

$$h_x = \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg} k_l^3}{4 \mu_l (T_{\text{sat}} - T_s) x} \right]^{1/4} \quad (10-20)$$

The average heat transfer coefficient over the entire plate is determined from its definition by substituting the h_x relation and performing the integration. It gives

$$h = h_{\text{vert}} = \frac{1}{L} \int_0^L h_x dx = \frac{4}{3} h_{x=L} = 0.943 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg} k_l^3}{\mu_l (T_{\text{sat}} - T_s) L} \right]^{1/4} \quad (10-21)$$

Equation 10–21, which is obtained with the simplifying assumptions stated earlier, provides good insight into the functional dependence of the condensation heat transfer coefficient. However, it is observed to underpredict heat transfer because it does not take into account the effects of the nonlinear temperature profile in the liquid film and the cooling of the liquid below the saturation temperature. Both of these effects can be accounted for by replacing h_{fg} with h_{fg}^* given by Eq. 10–9a. With this modification, the *average heat transfer coefficient* for laminar film condensation over a vertical flat plate of height L is determined to be

$$h_{\text{vert, laminar}} = 0.943 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) L} \right]^{1/4} \quad (\text{W/m}^2 \cdot \text{K}), \quad 0 < \text{Re} < 30 \quad (10-22)$$

where

- g = gravitational acceleration, m/s^2
- ρ_l, ρ_v = densities of the liquid and vapor, respectively, kg/m^3
- μ_l = viscosity of the liquid, $\text{kg/m} \cdot \text{s}$
- $h_{fg}^* = h_{fg} + 0.68 c_{pl} (T_{\text{sat}} - T_s)$ = modified latent heat of vaporization, J/kg
- k_l = thermal conductivity of the liquid, $\text{W/m} \cdot \text{K}$

L = height of the vertical plate, m

T_s = surface temperature of the plate, °C

T_{sat} = saturation temperature of the condensing fluid, °C

At a given temperature, $\rho_v \ll \rho_l$ and thus $\rho_l - \rho_v \approx \rho_l$ except near the critical point of the substance. Using this approximation and substituting Eqs. 10–14 and 10–18 at $x = L$ into Eq. 10–8 by noting that $\delta_{x=L} = k/h_{x=L}$ and $h_{\text{vert}} = \frac{4}{3}h_{x=L}$ (Eqs. 10–19 and 10–21) give

$$\text{Re}_{\text{vert, laminar}} = \frac{4g\rho_l(\rho_l - \rho_v)\delta^3}{3\mu_l^2} = \frac{4g\rho_l^2}{3\mu_l^2} \left(\frac{k_l}{h_{x=L}} \right)^3 = \frac{4g}{3\nu_l^2} \left(\frac{k_l}{3h_{\text{vert}}/4} \right)^3 \quad (10-23)$$

Then the heat transfer coefficient h_{vert} in terms of Re becomes

$$h_{\text{vert, laminar}} \cong 1.47k_l \text{Re}^{-1/3} \left(\frac{g}{\nu_l^2} \right)^{1/3}, \quad \begin{matrix} 0 < \text{Re} < 30 \\ \rho_v \ll \rho_l \end{matrix} \quad (10-24)$$

The results obtained from the theoretical relations above are in excellent agreement with the experimental results. It can be shown easily that using property values in Eqs. 10–22 and 10–24 in the *specified units* gives the condensation heat transfer coefficient in $\text{W/m}^2\cdot\text{K}$, thus saving one from having to go through tedious unit manipulations each time (Fig. 10–25). This is also true for the equations below. All properties of the liquid are to be evaluated at the film temperature $T_f = (T_{\text{sat}} + T_s)/2$. The h_{fg} and ρ_v are to be evaluated at the saturation temperature T_{sat} .

Wavy Laminar Flow on Vertical Plates

At Reynolds numbers greater than about 30, it is observed that waves form at the liquid–vapor interface, although the flow in liquid film remains laminar. The flow in this case is said to be *wavy laminar*. The waves at the liquid–vapor interface tend to increase heat transfer. But the waves also complicate the analysis and make it very difficult to obtain analytical solutions. Therefore, we have to rely on experimental studies. The increase in heat transfer due to the wave effect is, on average, about 20 percent, but it can exceed 50 percent. The exact amount of enhancement depends on the Reynolds number. Based on his experimental studies, Kutateladze (1963) recommended the following relation for the average heat transfer coefficient in wavy laminar condensate flow for $\rho_v \ll \rho_l$ and $30 < \text{Re} < 1800$,

$$h_{\text{vert, wavy}} = \frac{\text{Re} k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3}, \quad \begin{matrix} 30 < \text{Re} < 1800 \\ \rho_v \ll \rho_l \end{matrix} \quad (10-25)$$

A simpler alternative to the relation proposed by Kutateladze (1963) is

$$h_{\text{vert, wavy}} = 0.8 \text{Re}^{0.11} h_{\text{vert (smooth)}} \quad (10-26)$$

which relates the heat transfer coefficient in wavy laminar flow to that in wave-free laminar flow. McAdams (1954) went even further and suggested accounting for the increase in heat transfer in the wavy region by simply increasing the heat transfer coefficient determined from Eq. 10–22 for the laminar case by 20 percent. It is also suggested that Eq. 10–22 can be used for the wavy region, with the understanding that this is a conservative approach that provides a safety margin in thermal design. In this book we use Eq. 10–25.

$$\begin{aligned} h_{\text{vert}} &= \left(\frac{\frac{\text{m}}{\text{s}^2} \frac{\text{kg}}{\text{m}^3} \frac{\text{kg}}{\text{m}^3} \frac{\text{J}}{\text{kg}} \left(\frac{\text{W}}{\text{m} \cdot \text{K}} \right)^3}{\frac{\text{kg}}{\text{m} \cdot \text{s}} \cdot \text{K} \cdot \text{m}} \right)^{1/4} \\ &= \left[\frac{\text{m}}{\text{s}} \frac{1}{\text{m}^6} \frac{\text{W}^3}{\text{m}^3 \cdot \text{K}^3} \frac{\text{J}}{\text{K}} \right] \\ &= \left(\frac{\text{W}^4}{\text{m}^8 \cdot \text{K}^4} \right)^{1/4} \\ &= \text{W/m}^2 \cdot \text{K} \end{aligned}$$

FIGURE 10–25

Equation 10–22 gives the condensation heat transfer coefficient in $\text{W/m}^2\cdot\text{K}$ when the quantities are expressed in the units specified in their descriptions.

A relation for the Reynolds number in the wavy laminar region can be determined by substituting the h relation in Eq. 10–25 into the Re relation in Eq. 10–11 and simplifying. It yields

$$\text{Re}_{\text{vert, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820}, \quad \rho_v \ll \rho_l \quad (10-27)$$

Turbulent Flow on Vertical Plates

At a Reynolds number of about 1800, the condensate flow becomes turbulent. Several empirical relations of varying degrees of complexity are proposed for the heat transfer coefficient for turbulent flow. Again assuming $\rho_v \ll \rho_l$ for simplicity, Labuntsov (1957) proposed the following relation for the turbulent flow of condensate on *vertical plates*:

$$h_{\text{vert, turbulent}} = \frac{\text{Re } k_l}{8750 + 58 \text{Pr}_l^{0.5} (\text{Re}^{0.75} - 253)} \left(\frac{g}{\nu_l^2} \right)^{1/3}, \quad \text{Re} > 1800, \quad \rho_v \ll \rho_l \quad (10-28)$$

The physical properties of the condensate are again to be evaluated at the film temperature $T_f = (T_{\text{sat}} + T_s)/2$. The Re relation in this case is obtained by substituting the preceding h relation into the Re relation in Eq. 10–11, which gives

$$\text{Re}_{\text{vert, turbulent}} = \left[\frac{0.0690 L k_l \text{Pr}_l^{0.5} (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} - 151 \text{Pr}_l^{0.5} + 253 \right]^{4/3} \quad (10-29)$$

Nondimensionalized heat transfer coefficients for the wave-free laminar, wavy laminar, and turbulent flow of condensate on vertical plates are plotted in Fig. 10–26.

2 Inclined Plates

Equation 10–22 was developed for vertical plates, but it can also be used for laminar film condensation on the upper surfaces of plates that are *inclined* by an angle θ from the *vertical*, by replacing g in that equation with $g \cos \theta$

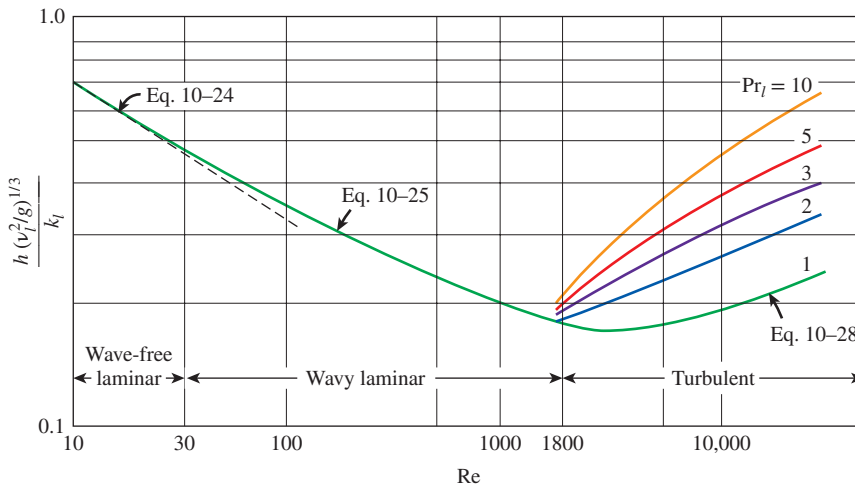


FIGURE 10–26

Nondimensionalized heat transfer coefficients for the wave-free laminar, wavy laminar, and turbulent flow of condensate on vertical plates.

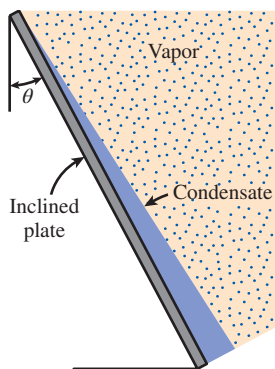


FIGURE 10-27
Film condensation on
an inclined plate.

(Fig. 10–27). This approximation gives satisfactory results, especially for $\theta \leq 60^\circ$. Note that the condensation heat transfer coefficients on vertical and inclined plates are related to each other by

$$h_{\text{inclined}} = h_{\text{vert}}(\cos \theta)^{1/4} \quad (\text{laminar}) \quad (10-30)$$

Equation 10–30 is developed for laminar flow of condensate, but it can also be used for wavy laminar flows as an approximation.

3 Vertical Tubes

Equation 10–22 for vertical plates can also be used to calculate the average heat transfer coefficient for laminar film condensation on the outer surfaces of vertical tubes, provided that the tube diameter is large relative to the thickness of the liquid film ($D \gg \delta$). In most of the problems encountered in this chapter, δ (the thickness of the liquid film at the bottom of the tube) can be calculated by combining Eqs. 10–19 and 10–21, which leads to $\delta = 4k_l/3h$.

4 Horizontal Tubes and Spheres

Nusselt's analysis of film condensation on vertical plates can also be extended to horizontal tubes and spheres. The average heat transfer coefficient for film condensation on the outer surfaces of a *horizontal tube* is determined to be

$$h_{\text{horiz}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \right]^{1/4} \quad (\text{W/m}^2 \cdot \text{K}) \quad (10-31)$$

where D is the diameter of the horizontal tube. Equation 10–31 can easily be modified for a *sphere* by replacing the constant 0.729 by 0.815.

A comparison of the heat transfer coefficient relations for a vertical tube of height L (Eq. 10–22) and a horizontal tube of diameter D (Eq. 10–31) yields

$$\frac{h_{\text{vert}}}{h_{\text{horiz}}} = \frac{0.943/L^{1/4}}{0.729/D^{1/4}} = 1.29 \left(\frac{D}{L} \right)^{1/4} \quad (10-32)$$

Setting $h_{\text{vert}} = h_{\text{horiz}}$ gives $L = 1.29^4 D = 2.77D$, which implies that for a tube whose length is 2.77 times its diameter, the average heat transfer coefficient for laminar film condensation will be the *same* whether the tube is positioned horizontally or vertically. For $L > 2.77D$, the heat transfer coefficient is higher in the horizontal position. Considering that the length of a tube in any practical application is several times its diameter, it is common practice to place the tubes in a condenser *horizontally* to maximize the condensation heat transfer coefficient on the outer surfaces of the tubes.

5 Horizontal Tube Banks

Horizontal tubes stacked on top of each other as shown in Fig. 10–28 are commonly used in condenser design. The average thickness of the liquid film at the lower tubes is much larger as a result of condensate falling on top of them from the tubes directly above. Therefore, the average heat transfer coefficient at the lower tubes in such arrangements is smaller. Assuming the condensate from the tubes above to the ones below drain smoothly, the average film condensation heat transfer coefficient for all tubes in a vertical tier can be expressed as

$$h_{\text{horiz, } N \text{ tubes}} = 0.729 \left[\frac{g\rho_l(\rho_l - \rho_v)h_{fg}^*k_l^3}{\mu_l(T_{\text{sat}} - T_s)ND} \right]^{1/4} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} \quad (10-33)$$

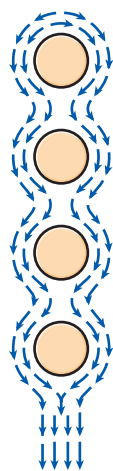


FIGURE 10-28
Film condensation on a vertical tier of
horizontal tubes.

Note that Eq. 10–33 can be obtained from the heat transfer coefficient relation for a horizontal tube (Eq. 10–31) by replacing D in that relation with ND , where N is the number of horizontal tubes in the tier. This relation does not account for the increase in heat transfer due to the ripple formation and turbulence caused during drainage, and thus generally yields conservative results.

Effect of Vapor Velocity

In the preceding analysis we assumed the vapor velocity to be small and thus the vapor drag exerted on the liquid film to be negligible, which is usually the case. However, when the vapor velocity is high, the vapor will “pull” the liquid at the interface along since the vapor velocity at the interface must drop to the value of the liquid velocity. If the vapor flows downward (i.e., in the same direction as the liquid), this additional force will increase the average velocity of the liquid and thus decrease the film thickness. This, in turn, will decrease the thermal resistance of the liquid film and thus increase heat transfer. Upward vapor flow has the opposite effects: the vapor exerts a force on the liquid in the opposite direction to flow, thickens the liquid film, and thus decreases heat transfer. Condensation in the presence of high vapor flow is studied (e.g., Shekriladze et al., 1966) and heat transfer relations are obtained, but a detailed analysis of this topic is beyond the scope of this introductory text.

The Presence of Noncondensable Gases in Condensers

Most condensers used in steam power plants operate at pressures well below the atmospheric pressure (usually under 0.1 atm) to maximize cycle thermal efficiency, and operation at such low pressures raises the possibility of air (a noncondensable gas) leaking into the condensers. Experimental studies show that the presence of noncondensable gases in the vapor has a detrimental effect on condensation heat transfer. Even small amounts of a noncondensable gas in the vapor cause significant drops in heat transfer coefficient during condensation. For example, the presence of less than 1 percent (by mass) of air in steam can reduce the condensation heat transfer coefficient by more than half. Therefore, it is common practice to periodically vent out the noncondensable gases that accumulate in the condensers to ensure proper operation.

The drastic reduction in the condensation heat transfer coefficient in the presence of a noncondensable gas can be explained as follows: When the vapor mixed with a noncondensable gas condenses, only the noncondensable gas remains in the vicinity of the surface (Fig. 10–29). This gas layer acts as a *barrier* between the vapor and the surface and makes it difficult for the vapor to reach the surface. The vapor now must diffuse through the noncondensable gas first before reaching the surface, and this reduces the effectiveness of the condensation process.

Experimental studies show that heat transfer in the presence of a noncondensable gas strongly depends on the nature of the vapor flow and the flow velocity. As you would expect, a *high flow velocity* is more likely to remove the stagnant noncondensable gas from the vicinity of the surface and thus *improve* heat transfer.

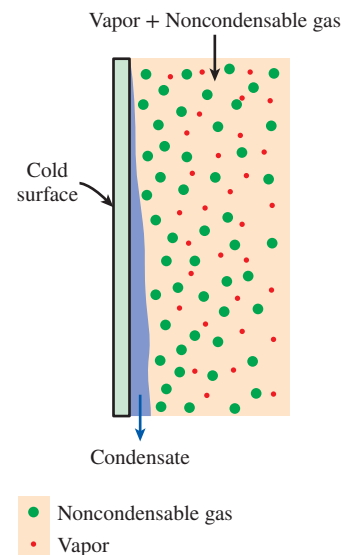


FIGURE 10–29

The presence of a noncondensable gas in a vapor prevents the vapor molecules from reaching the cold surface easily, and thus impedes condensation heat transfer.

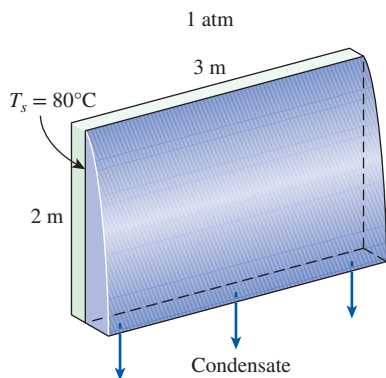


FIGURE 10–30
Schematic for Example 10–4.

EXAMPLE 10–4 Condensation of Steam on a Vertical Plate

Saturated steam at atmospheric pressure condenses on a 2-m-high and 3-m-wide vertical plate that is maintained at 80°C by circulating cooling water through the other side (Fig. 10–30). Determine (a) the rate of heat transfer by condensation to the plate and (b) the rate at which the condensate drips off the plate at the bottom.

SOLUTION Saturated steam at 1 atm condenses on a vertical plate. The rates of heat transfer and condensation are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The plate is isothermal. 3 The condensate flow is wavy-laminar over the entire plate (will be verified). 4 The density of vapor is much smaller than the density of liquid, $\rho_v \ll \rho_l$.

Properties The properties of water at the saturation temperature of 100°C are $h_{fg} = 2257 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.60 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (100 + 80)/2 = 90^\circ\text{C}$ are (Table A–9)

$$\begin{aligned}\rho_l &= 965.3 \text{ kg/m}^3 & c_{pl} &= 4206 \text{ J/kg}\cdot\text{K} \\ \mu_l &= 0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s} & k_l &= 0.675 \text{ W/m}\cdot\text{K} \\ \nu_l &= \mu_l/\rho_l = 0.326 \times 10^{-6} \text{ m}^2/\text{s}\end{aligned}$$

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned}h_{fg}^* &= h_{fg} + 0.68 c_{pl} (T_{\text{sat}} - T_s) \\ &= 2257 \times 10^3 \text{ J/kg} + 0.68 \times (4206 \text{ J/kg}\cdot\text{K})(100 - 80)^\circ\text{C} \\ &= 2314 \times 10^3 \text{ J/kg}\end{aligned}$$

For wavy-laminar flow, the Reynolds number is determined from Eq. 10–27 to be

$$\begin{aligned}\text{Re} &= \text{Re}_{\text{vert, wavy}} = \left[4.81 + \frac{3.70 L k_l (T_{\text{sat}} - T_s)}{\mu_l h_{fg}^*} \left(\frac{g}{\nu_l^2} \right)^{1/3} \right]^{0.820} \\ &= \left[4.81 + \frac{3.70(2 \text{ m})(0.675 \text{ W/m}\cdot\text{K})(100 - 80)^\circ\text{C}}{(0.315 \times 10^{-3} \text{ kg/m}\cdot\text{s})(2314 \times 10^3 \text{ J/kg})} \right. \\ &\quad \times \left. \left(\frac{9.81 \text{ m/s}^2}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} \right]^{0.820} \\ &= 1287\end{aligned}$$

which is between 30 and 1800, and thus our assumption of wavy laminar flow is verified. Then the condensation heat transfer coefficient is determined from Eq. 10–25 to be

$$\begin{aligned}h &= h_{\text{vert, wavy}} = \frac{\text{Re} k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3} \\ &= \frac{1287 \times (0.675 \text{ W/m}\cdot\text{K})}{1.08 (1287)^{1.22} - 5.2} \left(\frac{9.81 \text{ m/s}^2}{(0.326 \times 10^{-6} \text{ m}^2/\text{s})^2} \right)^{1/3} = 5850 \text{ W/m}^2\cdot\text{K}\end{aligned}$$

The heat transfer surface area of the plate is $A_s = W \times L = (3 \text{ m})(2 \text{ m}) = 6 \text{ m}^2$. Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (5850 \text{ W/m}^2\cdot\text{K})(6 \text{ m}^2)(100 - 80)^\circ\text{C} = \mathbf{7.02 \times 10^5 \text{ W}}$$

(b) The rate of condensation of steam is determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{7.02 \times 10^5 \text{ J/s}}{2314 \times 10^3 \text{ J/kg}} = \mathbf{0.303 \text{ kg/s}}$$

That is, steam will condense on the surface at a rate of 303 grams per second.

EXAMPLE 10-5 Condensation of Steam on a Tilted Plate

What would your answer be to the preceding example problem if the plate were tilted 30° from the vertical, as shown in Fig. 10–31?

SOLUTION (a) The heat transfer coefficient in this case can be determined from the vertical plate relation by replacing g with $g \cos \theta$. But we will use Eq. 10–30 instead since we already know the value for the vertical plate from the preceding example:

$$h = h_{\text{inclined}} = h_{\text{vert}} (\cos \theta)^{1/4} = (5850 \text{ W/m}^2 \cdot \text{K}) (\cos 30^\circ)^{1/4} = 5643 \text{ W/m}^2 \cdot \text{K}$$

The heat transfer surface area of the plate is still 6 m^2 . Then the rate of condensation heat transfer in the tilted plate case becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (5643 \text{ W/m}^2 \cdot \text{K}) (6 \text{ m}^2) (100 - 80)^\circ\text{C} = \mathbf{6.77 \times 10^5 \text{ W}}$$

(b) The rate of condensation of steam is again determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{6.77 \times 10^5 \text{ J/s}}{2314 \times 10^3 \text{ J/kg}} = \mathbf{0.293 \text{ kg/s}}$$

Discussion Note that the rate of condensation decreased by about 3.3 percent when the plate was tilted.

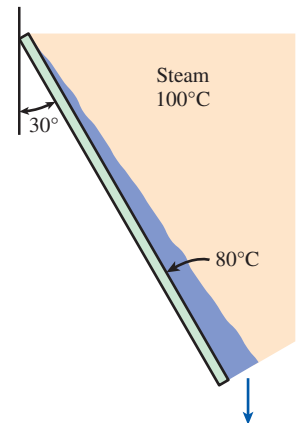


FIGURE 10-31

Schematic for Example 10-5.

EXAMPLE 10-6 Condensation of Steam on Horizontal Tubes

The condenser of a steam power plant operates at a pressure of 7.38 kPa. Steam at this pressure condenses on the outer surfaces of horizontal tubes through which cooling water circulates. The outer diameter of the pipes is 3 cm, and the outer surfaces of the tubes are maintained at 30°C (Fig. 10–32). Determine (a) the rate of heat transfer to the cooling water circulating in the tubes and (b) the rate of condensation of steam per unit length of a horizontal tube.

SOLUTION Saturated steam at a pressure of 7.38 kPa condenses on a horizontal tube at 30°C . The rates of heat transfer and condensation are to be determined.

Assumptions 1 Steady operating conditions exist. 2 The tube is isothermal.

Properties The properties of water at the saturation temperature of 40°C corresponding to 7.38 kPa are $h_{fg} = 2407 \times 10^3 \text{ J/kg}$ and $\rho_v = 0.05 \text{ kg/m}^3$. The properties of liquid water at the film temperature of $T_f = (T_{\text{sat}} + T_s)/2 = (40 + 30)/2 = 35^\circ\text{C}$ are (Table A–9)

$$\begin{aligned} \rho_l &= 994 \text{ kg/m}^3 & c_{pl} &= 4178 \text{ J/kg} \cdot \text{K} \\ \mu_l &= 0.720 \times 10^{-3} \text{ kg/m} \cdot \text{s} & k_l &= 0.623 \text{ W/m} \cdot \text{K} \end{aligned}$$

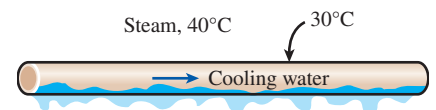


FIGURE 10-32

Schematic for Example 10-6.

Analysis (a) The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68 c_{pl}(T_{\text{sat}} - T_s) \\ &= 2407 \times 10^3 \text{ J/kg} + 0.68 \times (4178 \text{ J/kg}\cdot\text{K})(40 - 30)^\circ\text{C} \\ &= 2435 \times 10^3 \text{ J/kg} \end{aligned}$$

Noting that $\rho_v \ll \rho_l$ (since $0.05 \ll 994$), the heat transfer coefficient for condensation on a single horizontal tube is determined from Eq. 10–31 to be

$$\begin{aligned} h &= h_{\text{horiz}} = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \cong 0.729 \left[\frac{g \rho_l^2 h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4} \\ &= 0.729 \left[\frac{(9.81 \text{ m/s}^2)(994 \text{ kg/m}^3)^2 (2435 \times 10^3 \text{ J/kg})(0.623 \text{ W/m}\cdot\text{K})^3}{(0.720 \times 10^{-3} \text{ kg/m}\cdot\text{s})(40 - 30)^\circ\text{C}(0.03 \text{ m})} \right]^{1/4} \\ &= 9294 \text{ W/m}^2\cdot\text{K} \end{aligned}$$

The heat transfer surface area of the pipe per unit of its length is $A_s = \pi DL = \pi(0.03 \text{ m})(1 \text{ m}) = 0.09425 \text{ m}^2$. Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (9292 \text{ W/m}^2\cdot\text{K})(0.09425 \text{ m}^2)(40 - 30)^\circ\text{C} = \mathbf{8760 \text{ W}}$$

(b) The rate of condensation of steam is

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{8760 \text{ J/s}}{2435 \times 10^3 \text{ J/kg}} = \mathbf{0.00360 \text{ kg/s}}$$

Therefore, steam will condense on the horizontal tube at a rate of 3.6 g/s or 13.0 kg/h per meter of its length.

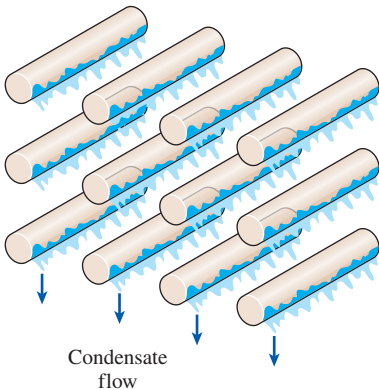


FIGURE 10–33

Schematic for Example 10–7.

EXAMPLE 10–7 Condensation of Steam on Horizontal Tube Banks

Repeat the preceding example problem for the case of 12 horizontal tubes arranged in a rectangular array of 3 tubes high and 4 tubes wide, as shown in Fig. 10–33.

SOLUTION (a) Condensation heat transfer on a tube is not influenced by the presence of other tubes in its neighborhood unless the condensate from other tubes drips on it. In our case, the horizontal tubes are arranged in four vertical tiers, each tier consisting of three tubes. The average heat transfer coefficient for a vertical tier of N horizontal tubes is related to the one for a single horizontal tube by Eq. 10–33 and is determined to be

$$h_{\text{horiz, } N \text{ tubes}} = \frac{1}{N^{1/4}} h_{\text{horiz, 1 tube}} = \frac{1}{3^{1/4}} (9294 \text{ W/m}^2\cdot\text{K}) = 7062 \text{ W/m}^2\cdot\text{K}$$

Each vertical tier consists of three tubes, and thus the heat transfer coefficient determined above is valid for each of the four tiers. In other words, this value can be taken to be the average heat transfer coefficient for all 12 tubes.

The surface area for all 12 tubes per unit length of the tubes is

$$A_s = N_{\text{total}} \pi D L = 12 \pi (0.03 \text{ m})(1 \text{ m}) = 1.1310 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = h A_s (T_{\text{sat}} - T_s) = (7062 \text{ W/m}^2 \cdot \text{K})(1.131 \text{ m}^2)(40 - 30)^\circ\text{C} = \mathbf{79,870 \text{ W}}$$

(b) The rate of condensation of steam is again determined from

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{79,870 \text{ J/s}}{2435 \times 10^3 \text{ J/kg}} = \mathbf{0.0328 \text{ kg/s}}$$

Therefore, steam will condense on the horizontal pipes at a rate of 32.8 g/s per meter length of the tubes.

10-6 ■ FILM CONDENSATION INSIDE HORIZONTAL TUBES

So far we have discussed film condensation on the *outer surfaces* of tubes and other geometries, which is characterized by negligible vapor velocity and the unrestricted flow of the condensate. Most condensation processes encountered in refrigeration and air-conditioning applications, however, involve condensation on the *inner surfaces* of horizontal or vertical tubes. Heat transfer analysis of condensation inside tubes is complicated by the fact that it is strongly influenced by the vapor velocity and the rate of liquid accumulation on the walls of the tubes (Fig. 10–34).

At high vapor velocities (Fig. 10–34a), the flow is characterized by two-phase annular conditions. The core of the annulus is occupied by the vapor, and its diameter decreases as the condensate thickness increases in the direction of the flow. At low vapor velocities (Fig. 10–34b), the condensate flow is from the upper portion of the tube to the bottom of the tube.

For *low vapor velocities*, Chato (1962) recommends this expression for condensation

$$h_{\text{internal}} = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \left(h_{fg} + \frac{3}{8} c_{pl} (T_{\text{sat}} - T_s) \right) \right]^{1/4} \quad (10-34)$$

for

$$\text{Re}_{\text{vapor}} = \left(\frac{\rho_v V_v D}{\mu_v} \right)_{\text{inlet}} < 35,000 \quad (10-35)$$

where the Reynolds number of the vapor is to be evaluated at the tube *inlet* conditions using the internal tube diameter as the characteristic length. Heat transfer coefficient correlations for higher vapor velocities are given by Rohsenow (1973).

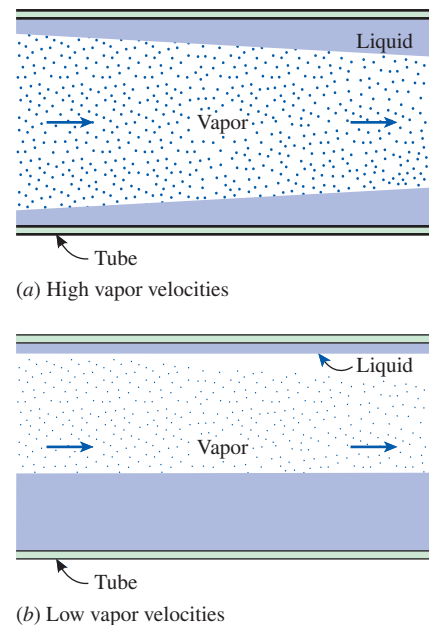


FIGURE 10-34

Condensate flow in a horizontal tube with high and low vapor velocities.

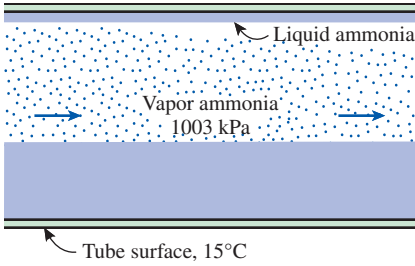


FIGURE 10–35

Schematic for Example 10–8.

PtD **EXAMPLE 10–8** Prevention of Ammonia Gas Release by Condensing It to Liquid

Exposure to high concentration of gaseous ammonia can cause respiratory tract and lung damage, and in some cases even death. One way of preventing the release of high concentration ammonia gas is storing it as liquid in pressurized vessels. Consider a process of condensing saturated ammonia vapor inside a horizontal tube. Saturated ammonia vapor enters a 25-mm-diameter tube with a length of 1.5 m, and it is condensed inside the tube at 1003 kPa. The surface temperature of the tube is maintained constant at 15°C (Fig. 10–35). To ensure that the ammonia is fully condensed to liquid at the exit of the 1.5-m-long tube, determine the flow rate of the ammonia entering the tube.

SOLUTION In this example, the concepts of Prevention through Design (PtD) are applied in conjunction with the concept of condensation inside a horizontal tube to condense vapor ammonia.

Assumptions 1 Steady operating conditions exist. 2 Tube is isothermal. 3 The vapor velocity is low so that $Re_{\text{vapor}} < 35,000$ (will be verified).

Properties The saturation temperature of ammonia at 1003 kPa is $T_{\text{sat}} = 25^\circ\text{C}$ (Table A–11). The properties of saturated ammonia at $T_{\text{sat}} = 25^\circ\text{C}$ are $h_{fg} = 1166 \times 10^3$ J/kg, $\rho_v = 7.809$ kg/m³, $\mu_v = 1.037 \times 10^{-5}$ kg/m·s. The properties of saturated ammonia at the film temperature of $T_f = (T_{\text{sat}} + T_s) = (25 + 15)/2 = 20^\circ\text{C}$ are

$$\rho_l = 610.2 \text{ kg/m}^3$$

$$\mu_l = 1.519 \times 10^{-4} \text{ kg/m}\cdot\text{s}$$

$$c_{pl} = 4745 \text{ J/kg}\cdot\text{K}$$

$$k_l = 0.4927 \text{ W/m}\cdot\text{K}$$

Analysis The heat transfer coefficient for condensation inside a horizontal tube is

$$\begin{aligned}
 h &= h_{\text{internal}} = 0.555 \left[\frac{g\rho_l(\rho_l - \rho_v)k_l^3}{\mu_l(T_{\text{sat}} - T_s)D} \left(h_{fg} + \frac{3}{8}c_{pl}(T_{\text{sat}} - T_s) \right) \right]^{1/4} \\
 h &= 0.555 \left[\frac{(9.81 \text{ m/s}^2)(610.2 \text{ kg/m}^3)(610.2 - 7.809) \text{ kg/m}^3 (0.4927 \text{ W/m}\cdot\text{K})^3}{(1.519 \times 10^{-4} \text{ kg/m}\cdot\text{s})(25 - 15) \text{ K}(0.025 \text{ m})} \right. \\
 &\quad \left. \times \left(1166 \times 10^3 \text{ J/kg} + \frac{3}{8}(4745 \text{ J/kg}\cdot\text{K})(25 - 15) \text{ K} \right) \right]^{1/4} \\
 &= 5976.3 \text{ W/m}^2\cdot\text{K}
 \end{aligned}$$

The heat transfer surface area of the tube is

$$A_s = \pi DL = \pi(0.025 \text{ m})(1.5 \text{ m}) = 0.11781 \text{ m}^2$$

Then the rate of heat transfer during this condensation process becomes

$$\dot{Q} = hA_s(T_{\text{sat}} - T_s) = (5976.3 \text{ W/m}^2\cdot\text{K})(0.11781 \text{ m}^2)(25 - 15) \text{ K} = 7040.7 \text{ W}$$

The modified latent heat of vaporization in this case is

$$\begin{aligned}
 h_{fg}^* &= h_{fg} + \frac{3}{8}c_{pl}(T_{\text{sat}} - T_s) \\
 &= 1166 \times 10^3 \text{ J/kg} + \frac{3}{8}(4745 \text{ J/kg}\cdot\text{K})(25 - 15) \text{ K} = 1183.8 \times 10^3 \text{ J/kg}
 \end{aligned}$$

Then, the rate of condensation becomes

$$\dot{m}_{\text{condensation}} = \frac{\dot{Q}}{h_{fg}^*} = \frac{7040.7 \text{ J/s}}{1183.8 \times 10^3 \text{ J/kg}} = 0.005948 \text{ kg/s}$$

Thus, $\dot{m} \leq 0.00595 \text{ kg/s}$

Discussion To ensure that the saturated ammonia vapor is fully condensed to liquid inside the 1.5-m-long tube, the flow rate of the ammonia entering the tube must be lower than the rate of condensation inside the tube.

Note that the low vapor velocity assumption is appropriate for $\dot{m} \leq 0.005948 \text{ kg/s}$, since

$$\begin{aligned} \text{Re}_{\text{vapor}} &= \left(\frac{\rho_v V_v D}{\mu_v} \right)_{\text{inlet}} = \frac{4\dot{m}}{\pi D \mu_v} = \frac{4(0.005948 \text{ kg/s})}{\pi(0.025 \text{ m})(1.037 \times 10^{-5} \text{ kg/m}\cdot\text{s})} \\ &= 29,212 < 35,000 \end{aligned}$$

10-7 ■ DROPWISE CONDENSATION

Dropwise condensation, characterized by countless droplets of varying diameters on the condensing surface instead of a continuous liquid film, is one of the most effective mechanisms of heat transfer, and extremely large heat transfer coefficients can be achieved with this mechanism (Fig. 10–36).

In dropwise condensation, the small droplets that form at the nucleation sites on the surface grow as a result of continued condensation, coalesce into large droplets, and slide down when they reach a certain size, clearing the surface and exposing it to vapor. There is no liquid film in this case to resist heat transfer. As a result, with dropwise condensation, heat transfer coefficients can be achieved that are more than 10 times larger than those associated with film condensation. Large heat transfer coefficients enable designers to achieve a specified heat transfer rate with a smaller surface area, and thus a smaller (and less expensive) condenser. Therefore, dropwise condensation is the preferred mode of condensation in heat transfer applications.

The challenge in dropwise condensation is not to achieve it, but rather, to *sustain* it for prolonged periods of time. Dropwise condensation is achieved by *adding* a promoting chemical into the vapor, *treating* the surface with a promoter chemical, or *coating* the surface with a polymer such as teflon or a noble metal such as gold, silver, rhodium, palladium, or platinum. The *promoters* used include various waxes and fatty acids such as oleic, stearic, and linoleic acids. They lose their effectiveness after a while, however, because of fouling, oxidation, and the removal of the promoter from the surface. It is possible to sustain dropwise condensation for over a year by the combined effects of surface coating and periodic injection of the promoter into the vapor. However, any gain in heat transfer must be weighed against the cost associated with sustaining dropwise condensation.

Dropwise condensation has been studied experimentally for a number of surface–fluid combinations. Of these, the studies on the condensation of steam on copper surfaces has attracted the most attention because of their



FIGURE 10–36

Dropwise condensation of water on a vertical surface.

©Chris Young/Alamy

widespread use in steam power plants. P. Griffith (1983) recommends these simple correlations for dropwise condensation of *steam* on *copper surfaces*:

$$h_{\text{dropwise}} = \begin{cases} 51,104 + 2044T_{\text{sat}}, & 22^\circ\text{C} < T_{\text{sat}} < 100^\circ\text{C} \\ 255,510 & T_{\text{sat}} > 100^\circ\text{C} \end{cases} \quad \begin{matrix} (10-36) \\ (10-37) \end{matrix}$$

where T_{sat} is in $^\circ\text{C}$ and the heat transfer coefficient h_{dropwise} is in $\text{W/m}^2\cdot\text{K}$.

The very high heat transfer coefficients achievable with dropwise condensation are of little significance if the material of the condensing surface is not a good conductor like copper or if the thermal resistance on the other side of the surface is too large. In steady operation, heat transfer from one medium to another depends on the sum of the thermal resistances on the path of heat flow, and a large thermal resistance may overshadow all others and dominate the heat transfer process. In such cases, improving the accuracy of a small resistance (such as one due to condensation or boiling) makes hardly any difference in overall heat transfer calculations.

EXAMPLE 10–9 Determination of Dropwise Condensation Rate

A can of beverage with a height of 123 mm and a diameter of 66 mm has a uniform temperature of 5°C when it is removed from the refrigerator. The can is placed on a table in a room and moist air condenses on the outer surface of the can (Fig. 10–37). Determine the dropwise condensation rate of the moisture in the air, if the saturation temperature is 25°C .

SOLUTION Beverage can at 5°C is placed in a room and condenses moist air at 25°C . The rate of dropwise condensation is to be determined.

Assumptions 1 Steady operating conditions exist. 2 The surface temperature of can is constant. 3 Condensation on top and bottom surface of the can is negligible.

Properties A required property of water at the saturation temperature $T_{\text{sat}} = 25^\circ\text{C}$ is $h_{fg} = 2442 \times 10^3 \text{ J/kg}$ (Table A–9). A required property of liquid water at the film temperature $T_f = (T_{\text{sat}} + T_s)/2 = (25 + 5)/2 = 15^\circ\text{C}$ is $c_{pl} = 4185 \text{ J/kg}\cdot\text{K}$ (Table A–9).

Analysis The modified latent heat of vaporization is

$$\begin{aligned} h_{fg}^* &= h_{fg} + 0.68 c_{pl} (T_{\text{sat}} - T_s) \\ &= 2442 \times 10^3 \text{ J/kg} + 0.68 (4185 \text{ J/kg}\cdot\text{K})(25 - 5) \text{ K} = 2.4989 \times 10^6 \text{ J/kg} \end{aligned}$$

The dropwise heat transfer coefficient, for $22^\circ\text{C} < T_{\text{sat}} < 100^\circ\text{C}$, is determined to be

$$\begin{aligned} h &= h_{\text{dropwise}} = 51,104 + 2044 T_{\text{sat}} \\ &= 51,104 + 2044(25^\circ\text{C}) = 102,204 \text{ W/m}^2\cdot\text{K} \end{aligned}$$

The rate of heat transfer to the beverage can during this condensation process is

$$\begin{aligned} \dot{Q} &= h A_s (T_{\text{sat}} - T_s) = h(\pi DL)(T_{\text{sat}} - T_s) \\ &= (102,204 \text{ W/m}^2\cdot\text{K}) \pi(0.066 \text{ m})(0.123 \text{ m})(25 - 5) \text{ K} = 5.2131 \times 10^4 \text{ W} \end{aligned}$$

Thus, the rate of condensation is

$$\dot{m} = \frac{\dot{Q}}{h_{fg}^*} = \frac{5.2131 \times 10^4 \text{ J/s}}{2.4989 \times 10^6 \text{ J/kg}} = \mathbf{0.02086 \text{ kg/s}}$$

Discussion The heat transfer rate during the dropwise condensation can be increased by increasing the surface area of the can, and thereby increasing the rate of condensation as well. Note that, if we had assumed film condensation in this case, the condensation rate estimate would be nearly 20 times less than the dropwise condensation.



FIGURE 10–37

Schematic for Example 10–9.

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TOPIC OF SPECIAL INTEREST*

Non-Boiling Two-Phase Flow Heat Transfer

The different flow patterns (regimes) encountered in flow boiling (see Fig. 10–19) can also be observed in non-boiling two-phase flow in both vertical and horizontal tubes (Figs. 10–38 to 10–40). Non-boiling two-phase flow has many industrial applications, such as the flow of oil and natural gas in flow lines and wellbores, in which the knowledge of non-boiling two-phase, two-component (liquid and permanent gas) heat transfer is required. During the production of two-phase hydrocarbon fluids from an oil reservoir to the surface, the temperature of the hydrocarbon fluids changes due to the difference in temperatures of the oil reservoir and the surface. The change in temperature results in heat transfer between the hydrocarbon fluids and the earth surrounding the oil well, and the ability to estimate the flowing temperature profile is necessary to address several design problems in petroleum production engineering.

In subsea oil and natural gas production, hydrocarbon fluids may leave the reservoir with a temperature of 75°C and flow in subsea surroundings of 4°C. As a result of the temperature gradient between the reservoir and the surroundings, the knowledge of heat transfer is critical to prevent gas hydrate and wax deposition blockages. Wax deposition can result in problems that include a reduction of inner tube diameter causing blockage, an increase in surface roughness of the tube leading to restricted flow line pressure, and a decrease in production, along with various mechanical problems.

In situations where low-velocity flow is necessary while high heat transfer rates are desirable, heat transfer enhancement schemes such as coil-spring wire insert, twisted tape insert, and helical ribs are used to promote turbulence, thus enhancing heat transfer. Although these heat transfer enhancement schemes have been proven to be effective, they do come with drawbacks, such as fouling, an increase in pressure drop, and sometimes even blockage. An alternative approach to enhance heat transfer in tube flow is by injecting gas into liquid to promote turbulence.

In the assessment of non-boiling heat transfer, void fraction (the ratio of the volume of space occupied by the gas phase to the total volume of the system) and flow patterns (qualitatively defined based on the distribution of one phase with respect to another phase across the pipe cross section) are two critical parameters that need to be taken into consideration. In any gas–liquid system, void fraction has great effects on heat transfer, since the liquid phase generally has significantly larger thermal conductivity than the gas phase. The knowledge of flow patterns in gas–liquid systems is also critical in the assessment of non-boiling heat transfer. For example, heat transfer in slug flow (bubbles coalesce into gas slugs) is likely to be different than heat transfer in annular flow (core of the flow consists of gas phase with entrained liquid droplets from the surrounding liquid phase). Void fraction, flow pattern, and hence the heat transfer are also affected by the tube inclination. Bhagwat and Ghajar (2012) reported that

*This section can be skipped without a loss of continuity.

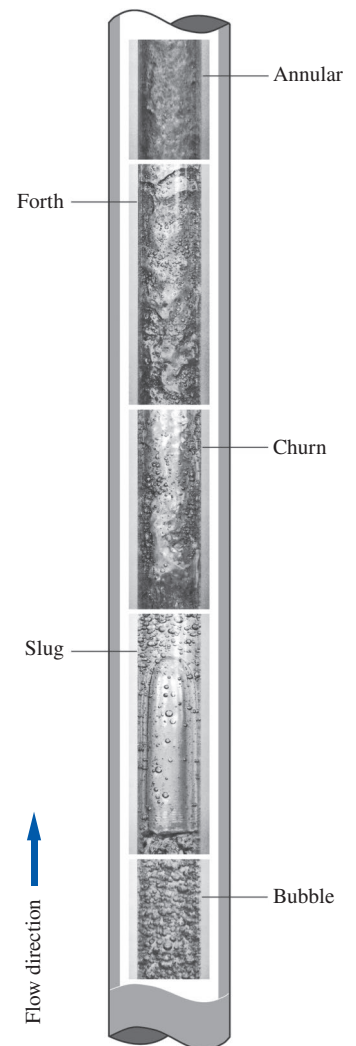
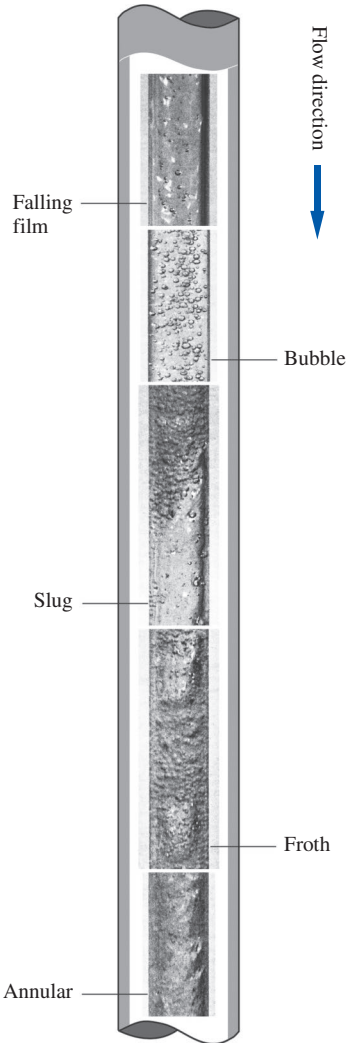


FIGURE 10–38

Flow patterns in upward vertical two-phase flow.

©Ghajar and Tang, 2009

**FIGURE 10-39**

Flow patterns in downward vertical two-phase flow.

©Ghajar and Tang, 2012

in inclined and vertical upward and downward tubes, the hydrodynamics of two-phase flow is influenced by the balance between buoyancy and gravitational force. The influence of buoyancy and gravitational force affects the slippage between the gas and liquid phases, which in turn affects the void fraction. Tube inclination also affects the flow patterns, which is the reason for the difference in flow patterns between vertical and horizontal flows (see Figs. 10–38 through 10–40). The influence of tube inclination in the upward and downward directions on the hydrodynamics of two-phase flow also leads to its effect on heat transfer as reported by Bhagwat and Ghajar (2016 and 2017). Due to the complexity and the lack of fundamental understanding of void fraction, flow patterns, and non-boiling heat transfer in two-phase flow, the majority of the documented works, whether they are flow pattern maps or correlations for void fraction and heat transfer, are only applicable to selected flow conditions. Ghajar and co-workers have extensively studied flow patterns, void fraction, pressure drop, and non-boiling heat transfer in gas–liquid two-phase flow in various pipe inclinations, and the results of their studies are summarized in the second edition of *Handbook of Multiphase Flow*; see Ghajar and Bhagwat (2017). Based on a database of 986 points, Ghajar and Tang (2009) recommended the following general correlation for the non-boiling two-phase heat transfer coefficient h_{tp} , for a variety of gas–liquid flow patterns, fluid combinations, and tube inclinations. The correlation captures 90 percent of the measured data within an error band of $\pm 25\%$ and 94 percent of the measured data within an error band of $\pm 30\%$,

$$h_{tp} = F_p h_l \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_p}{F_p} \right)^{0.4} \left(\frac{\text{Pr}_g}{\text{Pr}_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right], \text{ W/m}^2 \cdot \text{K} \quad (10-38)$$

where subscripts l and g refer to the liquid phase and gas phase, and

h_l = liquid phase heat transfer coefficient

$$= 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} (k_l/D) (\mu_l/\mu_s)^{0.14}, \text{ W/m}^2 \cdot \text{K}$$

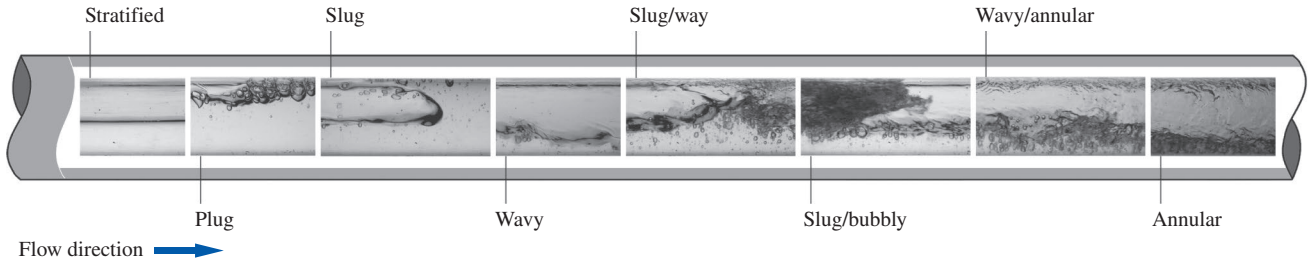
Re_l = in situ liquid Reynolds number $= 4\dot{m}_l/(\pi\sqrt{1-\alpha}\mu_l D)$

x = flow quality $= \dot{m}_g/(\dot{m}_g + \dot{m}_l)$

$$I^* = \text{inclination factor} = 1 + \frac{(\rho_l - \rho_g)gD^2}{\sigma} |\sin \theta|$$

$$F_p = \text{flow pattern factor} = (1 - \alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g(V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2$$

Here ρ is the density in kg/m^3 , V is the velocity in m/s , g is the gravitational acceleration in m/s^2 , D is the tube diameter in m , $\dot{m}$ is the mass flow rate in kg/s , μ is the dynamic viscosity in $\text{kg/m}\cdot\text{s}$, Pr is the Prandtl number, k is the thermal conductivity in $\text{W/m}\cdot\text{K}$, σ is the surface tension in N/m , and θ is the inclination angle in radians.

**FIGURE 10-40**

Flow patterns in horizontal two-phase flow.

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The void fraction α in Eq. 10–38 is to be determined from (Woldesemayat and Ghajar, 2007)

$$\alpha = \frac{V_{sg}}{C_o(V_{sg} + V_{sl}) + u_{gm}} \quad (10-39)$$

where C_o = two-phase distribution coefficient

$$= \frac{V_{sg}}{V_{sg} + V_{sl}} \left[1 + (V_{sl}/V_{sg})^{(\rho_s/\rho)^{0.1}} \right]$$

and u_{gm} = gas drift velocity

$$= 2.9(1.22 + 1.22 \sin \theta)^{P_{atm}/P_{sys}} \left[\frac{gD\sigma(1 + \cos \theta)(\rho_l - \rho_g)}{\rho_l^2} \right]^{0.25}, \text{ m/s}$$

Here P_{sys} and P_{atm} are the system and atmosphere pressures in N/m^2 , and V_{sl} and V_{sg} are the superficial liquid and gas velocities in m/s , respectively. In the preceding equations, all physical properties are evaluated at the *bulk mean temperature* $T_b = (T_i + T_e)/2$ except μ_s , which is evaluated at the surface temperature.

The non-boiling two-phase flow heat transfer correlation (Eq. 10–38) is based on 986 experimental data points (176 for horizontal flow, 555 for inclined flow, and 255 for upward vertical flow) and is applicable over the following range of parameters:

$$750 \leq \text{Re}_{sl} \leq 127,000, \quad 14 \leq \text{Re}_{sg} \leq 209,000,$$

$$9.99 \times 10^{-3} \leq \text{Pr}_g/\text{Pr}_l \leq 148 \times 10^{-3},$$

and

$$3.64 \times 10^{-3} \leq \mu_g/\mu_l \leq 26.3 \times 10^{-3}$$

where Re_{sl} and Re_{sg} are the superficial liquid and gas Reynolds numbers.

The application of the general two-phase flow non-boiling heat transfer correlation given above is illustrated next with a practical example problem dealing with a mixture of air and liquid silicone.

EXAMPLE 10–10 Air and Silicone (Dow Corning 200® Fluid, 5 cs) Flow in a Vertical Tube

Liquid silicone such as the Dow Corning 200® Fluid, 5 cs, is used primarily as an ingredient in cosmetic and personal care products due to its high spreadability, low surface tension ($\sigma = 19.7 \times 10^{-3}$ N/m), nongreasy, soft feel, and subtle skin lubricity characteristics. A two-phase flow of air and silicone (Dow Corning 200® Fluid, 5 cs) is being transported in an 11.7-mm-internal-diameter vertical tube. The liquid phase consists of liquid silicone with dynamic viscosity of 45.7×10^{-4} kg/m·s, density of 913 kg/m^3 , thermal conductivity of $0.117 \text{ W/m}\cdot\text{K}$, and Prandtl number of 64. The gas phase consists of air with dynamic viscosity of 18.4×10^{-6} kg/m·s, density of 1.19 kg/m^3 , and Prandtl number of 0.71. If the two-phase flow has flow quality of $x = 2.08 \times 10^{-5}$, void fraction of $\alpha = 0.011$, and liquid mass flow rate of $\dot{m}_l = 0.907 \text{ kg/s}$, estimate the two-phase heat transfer coefficient h_{tp} . Assume the dynamic viscosity of liquid silicone evaluated at the tube surface temperature to be $39.8 \times 10^{-4} \text{ kg/m}\cdot\text{s}$.

SOLUTION A two-phase flow of air and silicone (Dow Corning 200® Fluid, 5 cs) is being transported in an 11.7-mm-diameter vertical tube. Using the non-boiling two-phase heat transfer correlation, the two-phase heat transfer coefficient h_{tp} is to be determined.

Assumptions 1 Steady operating conditions exist. 2 Two-phase flow is non-boiling, and it does not involve phase change. 3 Fluid properties are constant.

Properties The properties of liquid silicone (liquid) are given to be $\mu_l = 45.7 \times 10^{-4} \text{ kg/m}\cdot\text{s}$, $\mu_s = 39.8 \times 10^{-4} \text{ kg/m}\cdot\text{s}$, $\rho_l = 913 \text{ kg/m}^3$, $k_l = 0.117 \text{ W/m}\cdot\text{K}$, $\sigma = 19.7 \times 10^{-3} \text{ N/m}$, and $\text{Pr}_l = 64$. The properties of air (gas) are given to be $\mu_g = 18.4 \times 10^{-6} \text{ kg/m}\cdot\text{s}$, $\rho_g = 1.19 \text{ kg/m}^3$, and $\text{Pr}_g = 0.71$.

Analysis With known liquid mass flow rate $\dot{m}_l$ and quality x , the gas mass flow rate $\dot{m}_g$ is determined using

$$\dot{m}_g = \frac{x}{1-x} \dot{m}_l = \frac{2.08 \times 10^{-5}}{1 - 2.08 \times 10^{-5}} (0.907 \text{ kg/s}) = 1.89 \times 10^{-5} \text{ kg/s}$$

From the gas and liquid mass flow rates, the superficial gas and liquid velocities can be calculated:

$$V_{sg} = \frac{\dot{m}_g}{\rho_g A} = \frac{4\dot{m}_g}{\rho_g \pi D^2} = \frac{4(1.89 \times 10^{-5} \text{ kg/s})}{(1.19 \text{ kg/m}^3)\pi(0.0117 \text{ m})^2} = 0.148 \text{ m/s}$$

$$V_{sl} = \frac{\dot{m}_l}{\rho_l A} = \frac{4\dot{m}_l}{\rho_l \pi D^2} = \frac{4(0.907 \text{ kg/s})}{(913 \text{ kg/m}^3)\pi(0.0117 \text{ m})^2} = 9.24 \text{ m/s}$$

Using the superficial velocities and void fraction, the gas and liquid velocities are found to be

$$V_g = \frac{V_{sg}}{\alpha} = \frac{0.148 \text{ m/s}}{0.011} = 13.5 \text{ m/s}$$

$$V_l = \frac{V_{sl}}{1-\alpha} = \frac{9.24 \text{ m/s}}{1-0.011} = 9.34 \text{ m/s}$$

The flow pattern factor (F_p) can be calculated using

$$\begin{aligned} F_p &= (1-\alpha) + \alpha \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{\rho_g(V_g - V_l)^2}{gD(\rho_l - \rho_g)}} \right) \right]^2 \\ &= (1-0.011) + (0.011) \left[\frac{2}{\pi} \tan^{-1} \left(\sqrt{\frac{(1.19 \text{ kg/m}^3)(13.5 \text{ m/s} - 9.34 \text{ m/s})^2}{(9.81 \text{ m/s}^2)(0.0117 \text{ m})(913 \text{ kg/m}^3 - 1.19 \text{ kg/m}^3)}} \right) \right]^2 \\ &= 0.9898 \end{aligned}$$

The inclination factor (I^*) for a vertical tube ($\theta = 90^\circ$) is calculated to be

$$\begin{aligned} I^* &= 1 + \frac{(\rho_l - \rho_g)gD^2}{\sigma} |\sin \theta| \\ &= 1 + \frac{(913 \text{ kg/m}^3 - 1.19 \text{ kg/m}^3)(9.81 \text{ m/s}^2)(0.0117 \text{ m})^2}{19.7 \times 10^{-3} \text{ N/m}} |\sin 90^\circ| \\ &= 63.16 \end{aligned}$$

The liquid phase heat transfer coefficient is calculated using:

$$\begin{aligned} h_l &= 0.027 \text{Re}_l^{4/5} \text{Pr}_l^{1/3} \left(\frac{k_l}{D} \right) \left(\frac{\mu_l}{\mu_s} \right)^{0.14} \\ &= 0.027 (21,718)^{4/5} (64)^{1/3} \left(\frac{0.117 \text{ W/m}\cdot\text{K}}{0.0117 \text{ m}} \right) \left(\frac{45.7 \times 10^{-4} \text{ kg/m}\cdot\text{s}}{39.8 \times 10^{-4} \text{ kg/m}\cdot\text{s}} \right)^{0.14} \\ &= 3246 \text{ W/m}^2\cdot\text{K} \end{aligned}$$

where the *in situ* liquid Reynolds number is

$$\begin{aligned} \text{Re}_l &= \frac{4\dot{m}_l}{\pi \sqrt{1-\alpha} \mu_l D} = \frac{4(0.907 \text{ kg/s})}{\pi \sqrt{1-0.011} (45.7 \times 10^{-4} \text{ kg/m}\cdot\text{s})(0.0117 \text{ m})} \\ &= 21,718 \end{aligned}$$

Using the general two-phase flow heat transfer correlation, Eq. 10–38, the two-phase heat transfer coefficient is estimated to be

$$\begin{aligned} \frac{h_{tp}}{h_l} &= F_p \left[1 + 0.55 \left(\frac{x}{1-x} \right)^{0.1} \left(\frac{1-F_p}{F_p} \right)^{0.4} \left(\frac{\text{Pr}_g}{\text{Pr}_l} \right)^{0.25} \left(\frac{\mu_l}{\mu_g} \right)^{0.25} (I^*)^{0.25} \right] \\ &= (0.9898) \left[1 + 0.55 \left(\frac{2.08 \times 10^{-5}}{1 - 2.08 \times 10^{-5}} \right)^{0.1} \left(\frac{1 - 0.9898}{0.9898} \right)^{0.4} \left(\frac{0.71}{64} \right)^{0.25} \right. \\ &\quad \times \left. \left(\frac{45.7 \times 10^{-4} \text{ kg/m}\cdot\text{s}}{18.4 \times 10^{-6} \text{ kg/m}\cdot\text{s}} \right)^{0.25} (63.16)^{0.25} \right] \\ &= 1.098 \end{aligned}$$

or

$$h_{tp} = 1.098 h_l = 1.098 (3246 \text{ W/m}^2\cdot\text{K}) = \mathbf{3560 \text{ W/m}^2\cdot\text{K}}$$

Discussion When compared with the two-phase heat transfer coefficient of $3480 \text{ W/m}^2\cdot\text{K}$ measured by Rezkallah (1987) in similar flow conditions, the general two-phase flow heat transfer correlation, Eq. 10–38, over-predicted the measured value by 2.3 percent.

Application of Reynolds Analogy to Non-Boiling Two-Phase Flow

We have seen in Chap. 6 that using the concept of the Reynolds analogy, the single-phase heat transfer coefficient can be predicted from a knowledge of the single-phase friction coefficient. Tang and Ghajar (2011) extended the application of the Reynolds analogy concept and developed a correlation to predict the non-boiling two-phase heat transfer coefficient from the knowledge of two-phase frictional pressure drop in horizontal and upward inclined flows. The correlation predicts more than 87 percent of the 637 data points within $\pm 30\%$ error bands. The correlation is expressed in following form,

$$\frac{h_{tp}}{h_l} = F_p^m \left(\frac{\dot{m}_l}{\dot{m}} \right) \left(\frac{\rho_{tp}}{\rho_l} \right)^n \phi_l^p \quad (10-40)$$

where the exponents m , n , and p have the values of 0.5, -0.5 , and 0.2 respectively, and

h_l = liquid phase heat transfer coefficient

$$= 0.027 \text{ Re}_l^{4/5} \text{Pr}_l^{1/3} (k_l/D) (\mu_l/\mu_s)^{0.14}, \text{ W/m}^2\cdot\text{K}$$

F_p = flow pattern factor same as that used in Eq. 10–38

ρ_{tp} = two-phase mixture density $= (1 - \alpha)\rho_l + \alpha\rho_g$, kg/m^3 . The void fraction (α) can be determined using Woldeesemayat and Ghajar (2007) correlation, Eq.10–39.

$$\phi_l = \text{two-phase frictional multiplier} = \sqrt{\frac{(dP/dL)_{f,tp}}{(dP/dL)_{f,l}}} \text{ (dimensionless quantity)}$$

$(dP/dL)_{f,l}$ and $(dP/dL)_{f,tp}$ are the single-phase liquid and two-phase frictional pressure drop, respectively. The friction factor required for the single-phase liquid pressure drop calculation can be obtained using the Moody chart (Fig. A–20) or the Colebrook equation (Eq. 8–76). If the two-phase frictional pressure drop is not known, then correlations recommended by Ghajar and Bhagwat (2014) may be used. The two-phase heat transfer correlation based on the Reynolds analogy (Eq. 10–40) was developed using experimental data for horizontal, upward inclined, and vertical upward pipe orientations using air–water and air–glycerin fluid combinations and is applicable over the following range of parameters:

For air–water: $700 \leq \text{Re}_{sl} \leq 107,000$, $14 \leq \text{Re}_{sg} \leq 150,000$

For air–glycerin: $50 \leq \text{Re}_{sl} \leq 83,000$, $40 \leq \text{Re}_{sg} \leq 59,000$

where Re_{sl} and Re_{sg} are the superficial liquid and gas Reynolds numbers, respectively.

For vertical downward flow, Bhagwat et al. (2012) pointed out that the two-phase heat transfer coefficient is less than that measured in vertical upward flow at similar mass flow rates, while the two-phase pressure drop is considerably higher in the former orientation than in the latter. Thus Eq. 10–40 with fixed exponents doesn't give satisfactory performance in the

prediction of the two-phase heat transfer coefficient for vertical downward flow. Bhagwat et al. (2012) proposed the following form of the Reynolds analogy-based correlation that can be used for vertical downward flow:

$$\frac{h_{tp}}{h_l} = \phi_l^{0.55} \quad (10-41)$$

The single-phase heat transfer coefficient (h_l) and the two-phase frictional multiplier (ϕ_l) are calculated in the same way as in Eq. 10-40.

Equation 10-41 is based on the 165 data points measured in a vertically downward oriented, 0.0127-m stainless steel pipe using an air–water fluid combination. This correlation predicts more than 90 percent of the measured data points within $\pm 20\%$ error bands and is applicable for $1400 \leq \text{Re}_{sl} \leq 29,000$ and $90 \leq \text{Re}_{sg} \leq 18,000$.

It should be noted that both Eqs. (10-38) and (10-40) require accurate information for the void fraction. Ghajar and Bhagwat (2013, 2014) have evaluated several void fraction correlations available in the literature against air–water data (5162 data points for horizontal, upward, and downward inclinations) and refrigerant–vapor (645 data points for horizontal flow). They found the performance of the Woldesemayat and Ghajar (2007) correlation or a more recent correlation by Bhagwat and Ghajar (2014) for a wider range of gas–liquid two-phase flow are acceptable for general application with requisite accuracy. For more specific flow conditions, other top-performing void fraction correlations shortlisted by Ghajar and Bhagwat (2013, 2014) may be used.

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SUMMARY

Boiling occurs when a liquid is in contact with a surface maintained at a temperature T_s sufficiently above the saturation temperature T_{sat} of the liquid. Boiling is classified as *pool boiling* or *flow boiling*, depending on the presence of bulk fluid motion. Boiling is called *pool boiling* in the absence of bulk fluid flow and *flow boiling* (or *forced convection boiling*) in its presence. Pool and flow boiling are further classified as *subcooled boiling* and *saturated boiling*, depending on the bulk liquid temperature. Boiling is said to be *subcooled* (or *local*) when the temperature of the main body of the liquid is below the saturation temperature T_{sat} and *saturated* (or *bulk*) when the temperature of the liquid is equal to T_{sat} . Boiling exhibits different regimes depending on the value of the excess temperature ΔT_{excess} . Four different boiling regimes are observed: natural convection boiling, nucleate boiling, transition boiling, and film boiling. These regimes are illustrated on the *boiling curve*. The rate of evaporation and the rate of heat transfer in nucleate boiling increase with

increasing ΔT_{excess} and reach a maximum at some point. The heat flux at this point is called the *critical* (or *maximum*) *heat flux*, $\dot{q}_{\text{max}}$. The rate of heat transfer in nucleate pool boiling is determined from

$$\dot{q}_{\text{nucleate}} = \mu_l h_{fg} \left[\frac{g(\rho_l - \rho_v)}{\sigma} \right]^{1/2} \left[\frac{c_{pl}(T_s - T_{\text{sat}})}{C_{sf} h_{fg} \text{Pr}_l^n} \right]^3$$

The *maximum* (or *critical*) *heat flux* in nucleate pool boiling is determined from

$$\dot{q}_{\text{max}} = C_{cr} h_{fg} [\sigma g \rho_v^2 (\rho_l - \rho_v)]^{1/4}$$

where the value of the constant C_{cr} is about 0.15. The minimum heat flux is given by

$$\dot{q}_{\text{min}} = 0.09 \rho_v h_{fg} \left[\frac{\sigma g (\rho_l - \rho_v)}{(\rho_l + \rho_v)^2} \right]^{1/4}$$

The heat flux for stable *film boiling* on the outside of a *horizontal cylinder* or *sphere* of diameter D is given by

$$\dot{q}_{\text{film}} = C_{\text{film}} \left[\frac{g k_v^3 \rho_v (\rho_l - \rho_v) h_{fg} + 0.4 c_{pv} (T_s - T_{\text{sat}})}{\mu_v D (T_s - T_{\text{sat}})} \right]^{1/4} \times (T_s - T_{\text{sat}})$$

where the constant $C_{\text{film}} = 0.62$ for horizontal cylinders and 0.67 for spheres. The *vapor* properties are to be evaluated at the *film temperature* $T_f = (T_{\text{sat}} + T_s)/2$, which is the *average temperature* of the vapor film. The liquid properties and h_{fg} are to be evaluated at the saturation temperature at the specified pressure.

Two distinct forms of condensation are observed in nature: film condensation and dropwise condensation. In *film condensation*, the condensate wets the surface and forms a liquid film on the surface that slides down under the influence of gravity. In *dropwise condensation*, the condensed vapor forms countless droplets of varying diameters on the surface instead of a continuous film.

The Reynolds number for the condensate flow is defined as

$$\text{Re} = \frac{D_h \rho_l V_l}{\mu_l} = \frac{4 A_c \rho_l V_l}{p \mu_l} = \frac{4 \dot{m}}{p \mu_l}$$

and

$$\text{Re} = \frac{4 \dot{Q}_{\text{conden}}}{p \mu_l h_{fg}^*} = \frac{4 A_s h (T_{\text{sat}} - T_s)}{p \mu_l h_{fg}^*}$$

where h_{fg}^* is the *modified latent heat of vaporization*, defined as

$$h_{fg}^* = h_{fg} + 0.68 c_{pl} (T_{\text{sat}} - T_s)$$

and represents heat transfer during condensation per unit mass of condensate.

Using some simplifying assumptions, the *average heat transfer coefficient* for smooth and wave-free laminar film condensation on a vertical plate of height L is determined to be

$$h_{\text{vert, laminar}} = 0.943 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) L} \right]^{1/4}$$

All properties of the *liquid* are to be evaluated at the film temperature $T_f = (T_{\text{sat}} + T_s)/2$. The h_{fg} and ρ_v are to be evaluated at T_{sat} . Condensate flow is *smooth* and *wave-free laminar* for about $\text{Re} \leq 30$, *wavy-laminar* in the range of $30 < \text{Re} < 1800$, and *turbulent* for $\text{Re} > 1800$. Heat transfer coefficients in the wavy-laminar and turbulent flow regions are determined from

$$h_{\text{vert, wavy}} = \frac{\text{Re } k_l}{1.08 \text{Re}^{1.22} - 5.2} \left(\frac{g}{\nu_l^2} \right)^{1/3}, \quad 30 < \text{Re} < 1800, \quad \rho_v \ll \rho_l$$

$$h_{\text{vert, turbulent}} = \frac{\text{Re } k_l}{8750 + 58 \text{Pr}^{-0.5} (\text{Re}^{0.75} - 253)} \left(\frac{g}{\nu_l^2} \right)^{1/3}, \quad \text{Re} > 1800, \quad \rho_v \ll \rho_l$$

Equations for vertical plates can also be used for laminar film condensation on the upper surfaces of the plates that are inclined by an angle θ from the vertical, by replacing g in that equation with $g \cos \theta$. Vertical plate equations can also be used to calculate the average heat transfer coefficient for laminar film condensation on the outer surfaces of vertical tubes, provided that the tube diameter is large relative to the thickness of the liquid film.

The average heat transfer coefficient for film condensation on the outer surfaces of a *horizontal tube* is determined to be

$$h_{\text{horiz}} = 0.729 \left[\frac{g \rho_l (\rho_l - \rho_v) h_{fg}^* k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \right]^{1/4}$$

where D is the diameter of the horizontal tube. This relation can easily be modified for a *sphere* by replacing the constant 0.729 with 0.815. It can also be used for N *horizontal tubes* stacked on top of each other by replacing D in the denominator with ND .

For low vapor velocities, film condensation heat transfer *inside horizontal tubes* can be determined from

$$h_{\text{internal}} = 0.555 \left[\frac{g \rho_l (\rho_l - \rho_v) k_l^3}{\mu_l (T_{\text{sat}} - T_s) D} \left(h_{fg} + \frac{3}{8} c_{pl} (T_{\text{sat}} - T_s) \right) \right]^{1/4}$$

$$\text{Re}_{\text{vapor}} = \left(\frac{\rho_v V_v D}{\mu_v} \right)_{\text{inlet}} < 35,000$$

where the Reynolds number of the vapor is to be evaluated at the tube inlet conditions using the internal tube diameter as the characteristic length. Finally, the heat transfer coefficient for *dropwise condensation* of steam on copper surfaces is given by

$$h_{\text{dropwise}} = \begin{cases} 51,104 + 2044 T_{\text{sat}}, & 22^\circ\text{C} < T_{\text{sat}} < 100^\circ\text{C} \\ 255,510, & T_{\text{sat}} > 100^\circ\text{C} \end{cases}$$

where T_{sat} is in $^\circ\text{C}$ and the heat transfer coefficient h_{dropwise} is in $\text{W/m}^2\cdot^\circ\text{C}$ or its equivalent $\text{W/m}^2\cdot\text{K}$.

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PROBLEMS*

Boiling Heat Transfer

10-1C What is boiling? What mechanisms are responsible for the very high heat transfer coefficients in nucleate boiling?

*Problems designated by a "C" are concept questions, and students are encouraged to answer them all. Problems designated by an "E" are in English units, and the SI users can ignore them. Problems with the  icon are comprehensive in nature and are intended to be solved with appropriate software. Problems with the  icon are Prevention through Design problems. Problems with the  icon are Codes and Standards problems.

10-2C Name the different boiling regimes in the order they occur in a vertical tube during flow boiling.

10-3C What is the difference between evaporation and boiling?

10-4C What is the difference between pool boiling and flow boiling?

10-5C Draw the boiling curve and identify the different boiling regimes. Also, explain the characteristics of each regime.

10-6C How does film boiling differ from nucleate boiling? Is the boiling heat flux necessarily higher in the stable film boiling regime than it is in the nucleate boiling regime?

10-7C Draw the boiling curve and identify the burnout point on the curve. Explain how burnout is caused. Why is the burnout point avoided in the design of boilers?

10-8C Discuss some methods of enhancing pool boiling heat transfer permanently.

10-9C Does the amount of heat absorbed as 1 kg of saturated liquid water boils at 100°C have to be equal to the amount of heat released as 1 kg of saturated water vapor condenses at 100°C?

10-10 Water is boiled at 120°C in a mechanically polished stainless-steel pressure cooker placed on top of a heating unit. The inner surface of the bottom of the cooker is maintained at 130°C. Determine the heat flux on the surface. *Answer: 228.4 kW/m²*

10-11 A platinum-plated rod with a diameter of 10 mm is submerged horizontally in water at atmospheric pressure. If the surface of the rod is maintained at 10°C above the saturation temperature, determine the nucleate pool boiling heat transfer rate per unit length and the rate of evaporation per unit length (in kg/s·m).

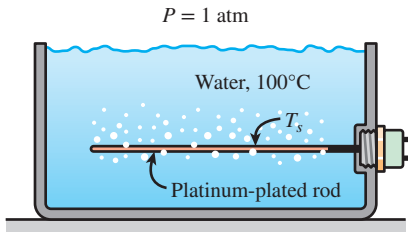


FIGURE P10-11

10-12 c & s Water is boiled at 361 kPa in a high-pressure system with an ASTM B371 brass rod. The ASTM B371 rod has a diameter and length of 5 mm and 15 cm, respectively. The rod is immersed in the water horizontally. The maximum use temperature for ASTM B371 brass rod is 149°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). The ASTM B371 rod is supplied with a heat transfer rate of 20 kW to boil the water. Determine if the supplied heat transfer rate would heat the rod surface temperature above the maximum use temperature.

10-13 A 65-cm-long, 2-cm-diameter brass heating element is to be used to boil water at 120°C. If the surface temperature of the heating element is not to exceed 125°C, determine the highest rate of steam production in the boiler, in kg/h. *Answer: 19.4 kg/h*

10-14 c & s An ASTM B75 copper tube sheathes a heating element that is used to boil water at 1254 kPa. The copper tube is immersed horizontally in the water, and its surface is polished. The tube diameter and length are 5 mm and 10 cm, respectively. The maximum use temperature for ASTM B75 copper tube is 204°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Determine the highest evaporation rate of water that can be achieved by the heater without heating the tube surface above the maximum use temperature.

10-15 Water is to be boiled at atmospheric pressure in a mechanically polished steel pan placed on top of a heating unit. The inner surface of the bottom of the pan is maintained at 110°C. If the diameter of the bottom of the pan is 30 cm, determine (a) the rate of heat transfer to the water and (b) the rate of evaporation.

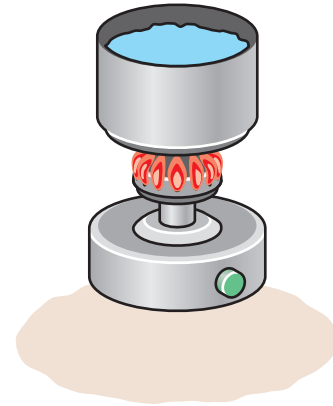


FIGURE P10-15

10–19 Water is boiled at atmospheric pressure by a horizontal platinum-plated rod with diameter of 10 mm. If the surface temperature of the rod is maintained at 110°C , determine the nucleate pool boiling heat transfer coefficient.

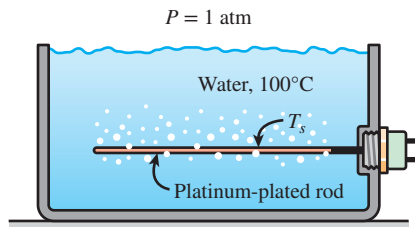


FIGURE P10–19

10–20 A 1-mm-diameter long electrical wire submerged in water at atmospheric pressure is dissipating 4100 W/m of heat, and the surface temperature reaches 128°C . If the experimental constant that depends on the fluid is $n = 1$, determine the nucleate boiling heat transfer coefficient and the value of the experimental constant C_{sf} .

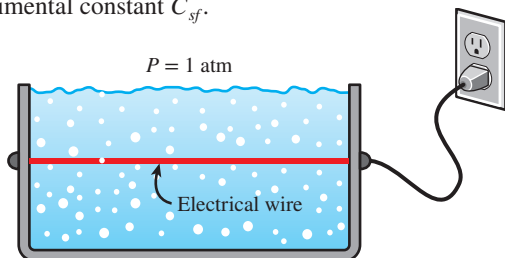


FIGURE P10–20

10–21 Water is boiled at sea level in a coffeemaker equipped with a 30-cm-long, 0.4-cm-diameter immersion-type electric heating element made of mechanically polished stainless steel. The coffeemaker initially contains 1 L of water at 14°C . Once boiling starts, it is observed that half of the water in the coffeemaker evaporates in 32 min. Determine the power rating of the electric heating element immersed in water and the surface temperature of the heating element. Also determine how long it will take for this heater to raise the temperature of 1 L of cold water from 14°C to the boiling temperature.

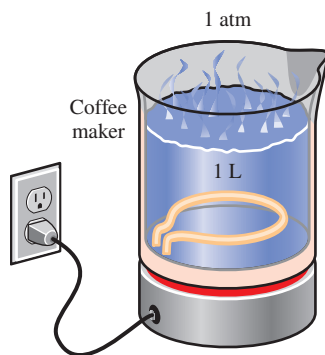


FIGURE P10–21

10–22 Repeat Prob. 10–21 for a copper heating element.

10–23 Water is boiled at 1 atm pressure in a 20-cm-internal-diameter Teflon®-pitted stainless steel pan on an electric range. If it is observed that the water level in the pan drops by 10 cm in 15 min, determine the inner surface temperature of the pan. *Answer:* 105.3°C

10–24 Repeat Prob. 10–23 for a polished copper pan.

10–25 Water is boiled at 120°C in a mechanically polished stainless steel pressure cooker placed on top of a heating unit. If the inner surface of the bottom of the cooker is maintained at 128°C , determine the boiling heat transfer coefficient.

10–26E Water is boiled at 250°F by a 2-ft-long and 0.25-in-diameter nickel-plated electric heating element maintained at 280°F . Determine (a) the boiling heat transfer coefficient, (b) the electric power consumed by the heating element, and (c) the rate of evaporation of water.

10–27E Repeat Prob. 10–26E for a platinum-plated heating element.

10–28E  Reconsider Prob. 10–26E. Using appropriate software, investigate the effect of surface temperature of the heating element on the boiling heat transfer coefficient, the electric power, and the rate of evaporation of water. Let the surface temperature vary from 260°F to 300°F . Plot the boiling heat transfer coefficient, the electric power consumption, and the rate of evaporation of water as a function of the surface temperature, and discuss the results.

10–29 Mechanically polished, 5-cm-diameter, stainless steel ball bearings are heated to 125°C uniformly. The ball bearings are then submerged in water at 1 atm to be cooled. Determine the rate of heat that is removed from a ball bearing at the instant it is submerged in the water.

10–30  A long, hot, mechanically polished stainless steel sheet ($c_p = 450\text{ J/kg}\cdot\text{K}$, $\rho = 7900\text{ kg/m}^3$) is being conveyed at 2 m/s through a water bath to be cooled. The 0.5-m-wide and 5-mm-thick stainless steel sheet has a temperature of 125°C as it enters the water bath. The length of the sheet submerged in water is 1 m as it is being conveyed through the water bath. As the hot sheet enters the water bath, boiling occurs at 1 atm. In order to prevent thermal burns to people handling the sheet, it must exit the water bath at a temperature below 45°C . Is the rate of heat that could be removed from the stainless steel sheet in the water bath sufficient to cool it to below 45°C as it leaves the water bath?

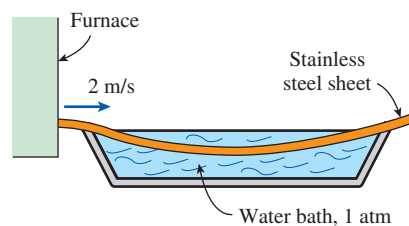


FIGURE P10–30

10–31 Water is boiled at 90°C by a horizontal brass heating element of diameter 8 mm. Determine the maximum heat flux that can be attained in the nucleate boiling regime.

10–32 A 1-mm-diameter nickel wire with electrical resistance of $0.129\ \Omega/\text{m}$ is submerged horizontally in water at atmospheric pressure. Determine the electrical current at which the wire would be in danger of burnout in nucleate boiling.

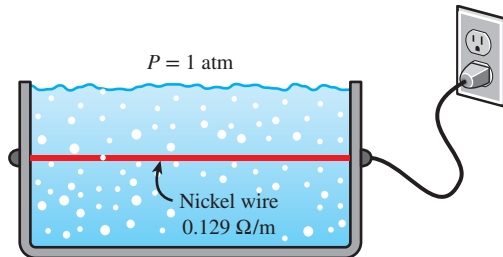


FIGURE P10–32

10–33 In a boiler, water is boiled at 100°C by hot gases flowing through a 10-m-long tube with an outer diameter of 5 cm that is submerged in the water. Determine the maximum rate of water vaporization that can be produced by the tube in the nucleate boiling regime.

10–34 Hot gases flow inside an array of tubes that are embedded in a 3-m $\times$ 3-m horizontal flat heater. The heater is used for boiling water in a tank at 160°C . In the interest of avoiding burnout, determine the maximum rate of water vaporization that can be achieved by the heater.

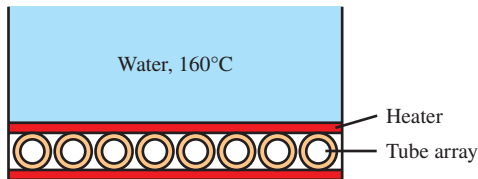


FIGURE P10–34

10–35 Water is boiled by two heaters of different geometries: a spherical heater and a horizontal flat heater. Both heaters are submerged in water that boils at 1 atm. The spherical heater has a diameter of 1 m, and the horizontal flat heater has the same surface area as the spherical heater. Determine the maximum rate of water vaporization that can be achieved by each heater without reaching burnout. Which heater geometry would produce a higher maximum rate of vaporization?

10–36 Hot gases flow inside an array of tubes that are embedded in a 3-m $\times$ 3-m horizontal flat heater. The heater surface is nickel-plated and is used for generating steam by boiling water in the nucleate boiling regime at 180°C . Determine the surface temperature of the heater and the convection heat transfer coefficient that is necessary to achieve the maximum rate of steam generation.

10–37 Water is to be boiled at atmospheric pressure on a 2-cm-diameter mechanically polished steel heater. Determine the maximum heat flux that can be attained in the nucleate boiling regime and the surface temperature of the heater surface in that case.

10–38  Reconsider Prob. 10–37. Using appropriate software, investigate the effect of local atmospheric pressure on the maximum heat flux and the temperature difference $T_s - T_{\text{sat}}$. Let the atmospheric pressure vary from 70 kPa to 101.3 kPa. Plot the maximum heat flux and the temperature difference as a function of the atmospheric pressure, and discuss the results.

10–39 A 10-cm $\times$ 10-cm horizontal flat heater is used for vaporizing refrigerant-134a at 350 kPa. The heater is supplied with $0.35\ \text{MW}/\text{m}^2$ of heat flux, and the surface temperature of the heater is 25°C . If the experimental constant in the Rohsenow correlation is $n = 1.7$, determine the value of the coefficient C_{sf} . Discuss whether or not the Rohsenow correlation is applicable in this analysis.

10–40 In a gas-fired boiler, water is boiled at 150°C by hot gases flowing through 25-m-long, 5-cm-outer-diameter mechanically polished stainless steel pipes submerged in water. If the outer surface temperature of the pipes is 160°C , determine (a) the rate of heat transfer from the hot gases to water, (b) the rate of evaporation, (c) the ratio of the critical heat flux to the present heat flux, and (d) the surface temperature of the pipe at which critical heat flux occurs.

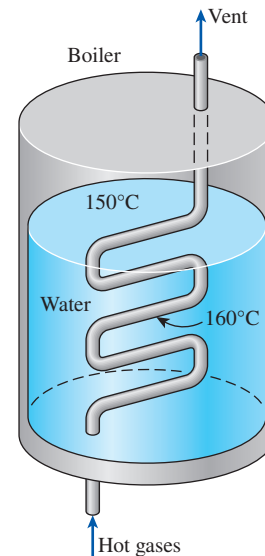


FIGURE P10–40

10–41 Repeat Prob. 10–40 for a boiling temperature of 155°C

10–42 A 1-m $\times$ 1-m horizontal flat heater is used for generating steam by boiling water at 1 atm. The boiling occurs at an excess temperature above 300°C , but it is maintained below

the burnout point. Determine the range of the steam generation rate that can be achieved by the heater.

10-43 Water is boiled at 90°C by a horizontal brass heating element of diameter 7 mm. Determine the surface temperature of the heater for the minimum heat flux case.

10-44  An ASTM B165 nickel–copper alloy tube sheathes a heating element that is used to boil water at 1 atm. The tube is immersed horizontally in the water. The tube diameter and length are 5 mm and 15 cm, respectively. The maximum use temperature for ASTM B165 nickel–copper alloy tube is 260°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Determine the highest evaporation rate of water that can be achieved by the heater without heating the tube surface above the maximum use temperature.

10-45 A 2-mm-diameter cylindrical metal rod with emissivity of 0.5 is submerged horizontally in water under atmospheric pressure. When electric current is passed through the metal rod, the surface temperature reaches 500°C . Determine the power dissipation per unit length of the metal rod.

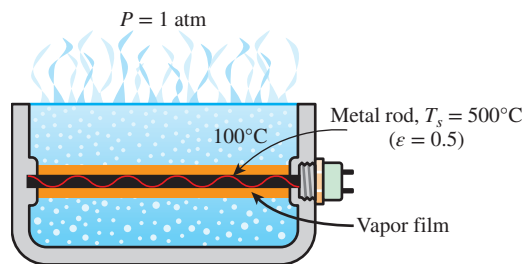


FIGURE P10-45

10-46E Water is boiled at atmospheric pressure by a horizontal polished copper heating element of diameter $D = 0.5$ in and emissivity $\epsilon = 0.2$ immersed in water. If the surface temperature of the heating element is 788°F , determine the rate of heat transfer to the water per unit length of the heating element.

10-47E Repeat Prob. 10-46E for a heating element temperature of 988°F .

10-48  An ASTM B335 nickel alloy rod, with a diameter of 5 mm and a length of 10 cm, is used to boil water at 1 atm. The rod is immersed in the water horizontally, and it has an emissivity of 0.3. The maximum use temperature for ASTM B335 nickel alloy rod is 427°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). Determine the highest rate of heat transfer that can be supplied from the ASTM B335 rod to boil the water without heating the rod surface above the maximum use temperature.

10-49 A 20-mm-diameter metal sphere has an emissivity of 0.75. The sphere was removed from a furnace when it reached a uniform temperature of 700°C and was suddenly submerged

in a water bath at atmospheric pressure. Estimate the initial heat transfer rate from the submerged metal sphere.

10-50 A 20-mm-diameter cylindrical steel rod with a length of 200 mm has an emissivity of 0.9. The steel rod was removed from a furnace when it reached a uniform temperature of 500°C and was suddenly submerged horizontally in a water bath at atmospheric pressure. Estimate the initial heat transfer rate from the submerged steel rod.

10-51 Water is boiled at 100°C by a spherical platinum heating element of diameter 15 cm and emissivity 0.10 immersed in the water. If the surface temperature of the heating element is 350°C , determine the rate of heat transfer from the heating element to the water.

10-52 Cylindrical stainless steel rods are heated to a uniform temperature of 700°C and then quenched in water at 1 atm. The rods are 25 cm in length, 25 mm in diameter, and their surface has an emissivity of 0.3. Determine the convection heat transfer coefficient and the total rate of heat removed from a rod at the instant it is submerged in the water during quenching.

10-53  A long cylindrical stainless steel rod ($c_p = 450$ J/kg·K, $\rho = 7900$ kg/m³, $\epsilon = 0.30$) with a mechanically polished surface is being conveyed through a water bath to be quenched. The 25-mm-diameter stainless steel rod has a temperature of 700°C as it enters the water bath. A length of 3 m of the rod is submerged in water as it is conveyed through the water bath during the quenching process. As the stainless steel rod enters the water bath, boiling occurs at 1 atm. In order to prevent thermal burns to people handling the rod, it must exit the water bath at a temperature below 45°C . Determine the speed of the rod being conveyed through the water bath so that it leaves the water bath without the risk of thermal burns.

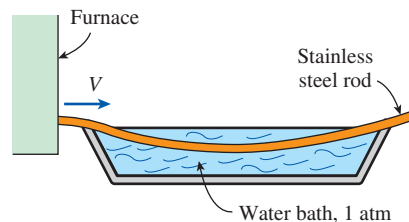


FIGURE P10-53

10-54 A cylindrical rod is used for boiling water at 1 atm. The rod has a diameter of 1 cm, and its surface has an emissivity of 0.3. Determine the film boiling convection heat transfer coefficient at the burnout point. Evaluate the properties of vapor at 1150°C . Discuss whether 1150°C is a reasonable film temperature for the vapor properties.

10-55 A 3-mm-diameter cylindrical heater is used for boiling water at 100°C . The heater surface is made of mechanically polished stainless steel with an emissivity of 0.3. Determine the boiling convection heat transfer coefficients at

the maximum heat flux for (a) nucleate boiling and (b) film boiling. For film boiling, evaluate the properties of vapor at 1150°C .

10-56 To understand the burnout phenomenon, boiling experiments are conducted in water at atmospheric pressure using an electrically heated 30-cm-long, 4-mm-diameter nickel-plated horizontal wire. Determine (a) the critical heat flux and (b) the increase in the temperature of the wire as the operating point jumps from the nucleate boiling to the film boiling regime at the critical heat flux. Take the emissivity of the wire to be 0.3.

10-57  Reconsider Prob. 10-56. Using appropriate software, investigate the effects of the local atmospheric pressure and the emissivity of the wire on the critical heat flux and the temperature rise of wire. Let the atmospheric pressure vary from 70 kPa to 101.3 kPa and the emissivity from 0.1 to 1.0. Plot the critical heat flux and the temperature rise as functions of the atmospheric pressure and the emissivity, and discuss the results.

Condensation Heat Transfer

10-58C What is condensation? How does it occur?

10-59C What is the difference between film and dropwise condensation? Which is a more effective mechanism of heat transfer?

10-60C How does the presence of a noncondensable gas in a vapor influence the condensation heat transfer?

10-61C What is the modified latent heat of vaporization? For what is it used? How does it differ from the ordinary latent heat of vaporization?

10-62C Consider film condensation on a vertical plate. Will the heat flux be higher at the top or at the bottom of the plate? Why?

10-63C Consider film condensation on the outer surfaces of four long tubes. For which orientation of the tubes will the condensation heat transfer coefficient be the highest: (a) vertical, (b) horizontal side by side, (c) horizontal but in a vertical tier (directly on top of each other), or (d) a horizontal stack of two tubes high and two tubes wide?

10-64C In condensate flow, how is the wetted perimeter defined? How does wetted perimeter differ from ordinary perimeter?

10-65 The Reynolds number for condensate flow is defined as $Re = 4\dot{m}/p\mu_f$, where p is the wetted perimeter. Obtain simplified relations for the Reynolds number by expressing p and $\dot{m}$ by their equivalence for the following geometries: (a) a vertical plate of height L and width w , (b) a tilted plate of height L and width w inclined at an angle u from the vertical, (c) a vertical cylinder of length L and diameter D , (d) a horizontal cylinder of length L and diameter D , and (e) a sphere of diameter D .

10-66 A vertical $0.2\text{-m} \times 0.2\text{-m}$ square plate is exposed to saturated water vapor at atmospheric pressure. If the surface temperature is 80°C and the flow is laminar, estimate the local heat transfer coefficients at the middle and at the bottom of the plate.

10-67 A vertical $0.5\text{-m} \times 0.5\text{-m}$ square plate is used in a process to condense saturated water vapor. If the desired rate of condensation is 0.016 kg/s , determine the necessary surface temperature of the plate at atmospheric pressure. For this problem, as a first approximation, assume a film temperature of 90°C for the evaluation of the liquid properties and a surface temperature of 80°C for the evaluation of modified latent heat of vaporization. Are these good assumptions?

10-68 Saturated ammonia vapor at 30°C is passed over 30 vertical flat plates, each of which is 10 cm high and 25 cm wide. The average surface temperature of the plates is 10°C . Estimate the average heat transfer coefficient and the rate of ammonia condensation. Assume wavy-laminar flow. Is this a good assumption?

10-69 Saturated steam at 1 atm condenses on a 2-m-high and 10-m-wide vertical plate that is maintained at 90°C by circulating cooling water through the other side. Determine (a) the rate of heat transfer by condensation to the plate, and (b) the rate at which the condensate drips off the plate at the bottom. Assume wavy-laminar flow. Is this a good assumption?

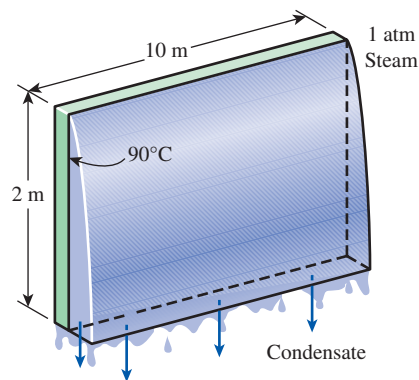


FIGURE P10-69

10-70 Repeat Prob. 10-69 for the case of the plate being tilted 60° from the vertical.

10-71  Saturated ammonia vapor in a pressurized system undergoes condensation at 190 kPa on a vertical plate. The plate is made of ASTM A240 410S stainless steel; it has a height of 2 m and a width of 1.5 m. According to the ASME Code for Process Piping, the minimum temperature suitable for ASTM A240 410S stainless steel plate is -30°C (ASME B31.3-2014, Table A-1M). Determine the highest rate of condensation that can be produced by the plate without cooling the plate below the minimum suitable

temperature set by the ASME Code for Process Piping. Assume wavy-laminar flow. Is this a good assumption?

10-72 Saturated steam at 100°C condenses on a $2\text{-m} \times 2\text{-m}$ plate that is tilted 30° from the vertical. The plate is maintained at 80°C by cooling it from the other side. Determine (a) the average heat transfer coefficient over the entire plate and (b) the rate at which the condensate drips off the plate at the bottom. Assume wavy-laminar flow. Is this a good assumption?

10-73  Reconsider Prob. 10-72. Using appropriate software, investigate the effects of plate temperature and the angle of the plate from the vertical on the average heat transfer coefficient and the rate at which the condensate drips off. Let the plate temperature vary from 40°C to 90°C and the plate angle from 0° to 60° . Plot the heat transfer coefficient and the rate at which the condensate drips off as functions of the plate temperature and the tilt angle, and discuss the results.

10-74 Saturated steam at 30°C condenses on the outside of a 4-cm-outer-diameter, 2-m-long vertical tube. The temperature of the tube is maintained at 20°C by the cooling water. Determine (a) the rate of heat transfer from the steam to the cooling water, (b) the rate of condensation of steam, and (c) the approximate thickness of the liquid film at the bottom of the tube. Assume wavy-laminar flow and that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube. Are these good assumptions?

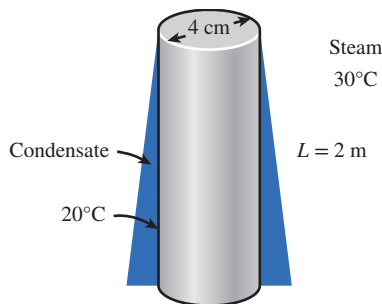


FIGURE P10-74

10-75 Saturated water vapor at atmospheric pressure condenses on the outer surface of a 0.1-m-diameter vertical pipe. The pipe is 1 m long and has a uniform surface temperature of 80°C . Determine the rate of condensation and the heat transfer rate by condensation. Discuss whether the pipe can be treated as a vertical plate. Assume wavy-laminar flow and that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube. Are these good assumptions?

10-76 Saturated steam at 55°C is to be condensed at a rate of 10 kg/h on the outside of a 3-cm-outer-diameter vertical tube whose surface is maintained at 45°C by the cooling water. Determine the required tube length. Assume wavy-laminar flow and that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube. Are these good assumptions?

10-77 Repeat Prob. 10-76 for a horizontal tube. *Answer: 0.70 m*

10-78 A 1.5-m-long vertical tube is used for condensing saturated steam at 60°C . The surface temperature of the tube is maintained at a uniform temperature of 40°C by flowing coolant inside the tube. Determine the heat transfer rate to the tube and the required tube diameter to condense 12 kg/h of steam during the condensation process. Assume wavy-laminar flow and that the tube diameter is large relative to the thickness of the liquid film at the bottom of the tube. Are these good assumptions?

10-79 Repeat Prob. 10-78 for a horizontal tube. Compare the required diameters between a vertical tube and a horizontal tube to condense 12 kg/h of steam during the condensation process.

10-80 Saturated ammonia vapor at 10°C condenses on the outside of a 4-cm-outer-diameter, 15-m-long horizontal tube whose outer surface is maintained at -10°C . Determine (a) the rate of heat transfer from the ammonia and (b) the rate of condensation of ammonia.

10-81  Saturated refrigerant-134a vapor undergoes condensation on the outer surface of a horizontal ASTM A268 TP443 stainless steel tube at 133 kPa. The tube is 50 cm long and has a diameter of 15 mm. According to the ASME Code for Process Piping, the minimum temperature suitable for ASTM A268 TP443 stainless steel tube is -30°C (ASME B31.3-2014, Table A-1M). Determine whether the condensation of refrigerant-134a on the tube surface at a rate of 3 g/s would cool the surface below the minimum suitable temperature set by the ASME Code for Process Piping.

10-82  A 3-m spherical tank storing cold fluid has a uniform surface temperature of 5°C . The lower temperature of the tank surface is causing condensation of moisture in the air at 25°C . The tank is situated in the vicinity of high-voltage devices, and water from the condensation on the tank surface can create an electrical hazard. To prevent an accident, a system is to be implemented to remove 0.5 kg/s of condensate from coming in contact with the high-voltage devices. Determine whether or not the preventive system can handle the rate of condensation from the tank surface.

10-83 Consider film condensation on the outer surfaces of N horizontal tubes arranged in a vertical tier. For what value of N will the average heat transfer coefficient for the entire stack of tubes be equal to half of what it is for a single horizontal tube?

10-84 A large heat exchanger has several columns of tubes, with 33 tubes in each column. The outer diameter of the tubes is 1.5 cm. Saturated steam at 50°C condenses on the outer surfaces of the tubes, which are maintained at 20°C . Determine (a) the average heat transfer coefficient and (b) the rate of condensation of steam per m length of a column.

10-85 The condenser of a steam power plant operates at a pressure of 4.25 kPa. The condenser consists of 144 horizontal

tubes arranged in a 12×12 square array. The tubes are 8 m long and have an outer diameter of 3 cm. If the tube surfaces are at 20°C , determine (a) the rate of heat transfer from the steam to the cooling water and (b) the rate of condensation of steam in the condenser. *Answers: (a) 5060 kW, (b) 2.06 kg/s*

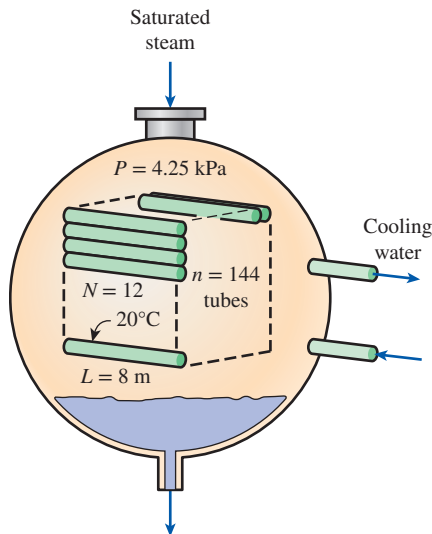


FIGURE P10-85

10-86  Reconsider Prob. 10-85. Using appropriate software, investigate the effect of the condenser pressure on the rate of heat transfer and the rate of condensation of the steam. Let the condenser pressure vary from 3 kPa to 15 kPa. Plot the rate of heat transfer and the rate of condensation of the steam as a function of the condenser pressure, and discuss the results.

10-87E Saturated steam at 95°F is condensed on the outer surfaces of an array of horizontal pipes through which cooling water circulates. The outer diameter of the pipes is 0.8 in and the outer surfaces of the pipes are maintained at 65°F . Determine (a) the rate of heat transfer to the cooling water circulating in the pipes and (b) the rate of condensation of steam per unit length of a single horizontal pipe.

10-88E Repeat Prob. 10-87E for the case of 32 horizontal pipes arranged in a rectangular array of 4 pipes high and 8 pipes wide.

10-89 A horizontal condenser uses a 4×4 array of tubes that have an outer diameter of 5.0 cm and length 2.0 m. Saturated steam at 101.3 kPa condenses on the outside tube surface held at a temperature of 80°C . Calculate the steady rate of steam condensation in kg/h.

10-90  Four long ASTM A437 B4B stainless steel bolts are used to hold two separated plates together. The bolts are cylindrical, and each has a diameter of 13 mm. Between the two plates, the horizontal bolts are exposed to

saturated propane vapor. The length of each bolt between the plates is 15 cm. The bolts are arranged in a vertical tier, and condensation of saturated propane occurs on the bolts at 344 kPa. The minimum temperature suitable for ASTM A437 B4B stainless steel bolts is -30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-2M). Determine the highest rate of condensation that can occur on the bolts, without cooling the bolts below the minimum suitable temperature set by the ASME Code for Process Piping.

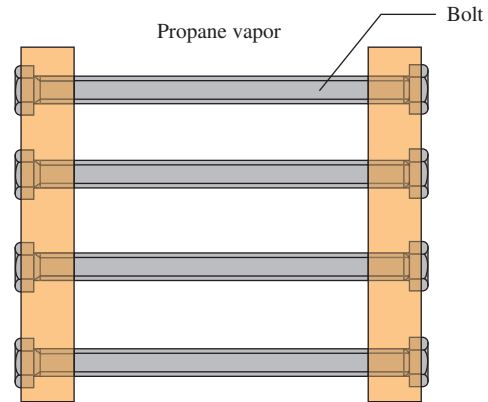


FIGURE P10-90

10-91 Saturated water vapor at a pressure of 12.4 kPa is condensed over 100 horizontal tubes in a rectangular array of 5 tubes high and 20 tubes wide, each with a diameter of 8 mm. If the tube surfaces are maintained with a uniform temperature of 30°C , determine the condensation rate per unit length (in kg/s·m) of the tubes.

10-92 Saturated water vapor at a pressure of 12.4 kPa is condensed over an array of 100 horizontal tubes, each with a diameter of 8 mm and a length of 1 m. The tube surfaces are maintained with a uniform temperature of 30°C . Determine the condensation rates of the tubes for (a) a rectangular array of 5 tubes high and 20 tubes wide and (b) a square array of 10 tubes high and 10 tubes wide. Compare and discuss the results of (a) and (b).

10-93 Saturated refrigerant-134a vapor at a pressure of 888 kPa is condensed as it is flowing through a 0.25-m-long tube with a diameter of 12 mm. The tube wall temperature is kept uniform at 15°C . If the vapor enters the tube at a flow rate of 0.003 kg/s, determine the flow rate of the vapor at the exit.

10-94  Saturated ammonia vapor flows inside a horizontal tube with a length and an inner diameter of 1 m and 25 mm, respectively. The tube is made of ASTM A268 TP443 stainless steel. The ammonia vapor enters the tube at a flow rate of 5 g/s, and condensation occurs inside the tube at 190 kPa. The minimum temperature suitable for ASTM A268 TP443 stainless steel tube is -30°C (ASME Code for Process Piping, ASME B31.3-2014, Table A-1M). If the flow

rate of ammonia vapor at the tube exit is 0.5 g/s, would the temperature of the tube wall comply with the ASME Code?

10-95 Saturated ammonia vapor at a pressure of 1003 kPa is condensed as it flows through a 25-mm tube. The tube length is 0.5 m, and the wall temperature is kept uniform at 5°C. If the vapor exits the tube at a flow rate of 0.002 kg/s, determine the flow rate of the vapor at the inlet. Assume the Reynolds number of the vapor at the tube inlet is less than 35,000. Is this a good assumption?

10-96 A 10-m-long copper tube with a diameter of 25 mm is used for transporting cold coolant. The coolant flow inside the tube maintains the tube surface temperature at 5°C and causes moist air to condense on the outer surface. Determine the dropwise condensation rate of the moisture in the air if the saturation temperature is 25°C.

10-97  A 10-m-long copper tube with a diameter of 25 mm is used for transporting cold fluid. The cold fluid flow inside the tube causes the tube surface temperature at 3°C to be colder than the ambient temperature. As a result, moist air condenses at 27°C on the tube outer surface. The tube is situated in the vicinity of high-voltage devices, and water from the condensation can cause an electrical hazard. It has been determined that both film condensation and dropwise condensation can occur on the tube surface. To prevent an electrical hazard, a system is to be implemented to remove the condensate and prevent it from coming in contact with the high-voltage devices. Size the system so that it is capable of handling the condensation rate from the tube surface.

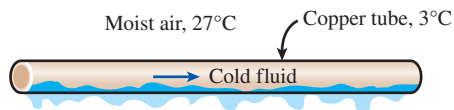


FIGURE P10-97

10-98 Steam is being condensed at 60°C by a 15-m-long horizontal copper tube with a diameter of 25 mm. The tube surface temperature is maintained at 40°C. Determine the condensation rate of the steam during (a) film condensation, and (b) dropwise condensation. Compare and discuss the results.

Special Topic: Non-Boiling Two-Phase Flow Heat Transfer

10-99 Consider a non-boiling gas–liquid two-phase flow in a tube, where the ratio of the mass flow rate is $\dot{m}_l/\dot{m}_g = 300$. Determine the flow quality (x) of this non-boiling two-phase flow.

10-100 Consider a non-boiling gas–liquid two-phase flow in a 102-mm diameter tube, where the superficial gas velocity is one-third that of the liquid. If the densities of the gas and liquid are $\rho_g = 8.5 \text{ kg/m}^3$ and $\rho_l = 855 \text{ kg/m}^3$, respectively, determine the flow quality and the mass flow rates of the gas and the liquid when the gas superficial velocity is 0.8 m/s.

10-101 A non-boiling two-phase flow of air and engine oil in a 25-mm-diameter tube has a bulk mean temperature of 140°C. If the flow quality is 2.1×10^{-3} and the mass flow rate of the engine oil is 0.9 kg/s, determine the mass flow rate of air and the superficial velocities of air and engine oil.

10-102 Consider a mixture of air–water flow in a tube. Suddenly the air flow is shut off, so that only water is flowing in the tube. Starting with the two-phase non-boiling heat transfer correlation, find the expression that is appropriate for this situation.

10-103 An air–water slug flows through a 25.4-mm-diameter horizontal tube in microgravity conditions (less than 1 percent of earth's normal gravity). The liquid phase consists of water with dynamic viscosity of $\mu_l = 85.5 \times 10^{-5} \text{ kg/m}\cdot\text{s}$, density of $\rho_l = 997 \text{ kg/m}^3$, thermal conductivity of $k_l = 0.613 \text{ W/m}\cdot\text{K}$, and Prandtl number of $Pr_l = 5.0$. The gas phase consists of air with dynamic viscosity of $\mu_g = 18.5 \times 10^{-6} \text{ kg/m}\cdot\text{s}$, density $\rho_g = 1.16 \text{ kg/m}^3$, and Prandtl number $Pr_g = 0.71$. At a superficial gas velocity of $V_{sg} = 0.3 \text{ m/s}$, a superficial liquid velocity of $V_{sl} = 0.544 \text{ m/s}$, and a void fraction of $\alpha = 0.27$, estimate the two-phase heat transfer coefficient h_{tp} . Assume the dynamic viscosity of water evaluated at the tube surface temperature to be $73.9 \times 10^{-5} \text{ kg/m}\cdot\text{s}$.

10-104 A two-phase flow mixture of air and silicone (Dow Corning 200® Fluid, 5 cs) is being transported in an 11.7-mm-diameter thin-walled horizontal tube. The liquid phase consists of liquid silicone with a surface tension (σ) of $19.7 \times 10^{-3} \text{ N/m}$, dynamic viscosity (μ_l) of $45.7 \times 10^{-4} \text{ kg/m}\cdot\text{s}$, density (ρ_l) of 913 kg/m^3 , thermal conductivity (k_l) of $0.117 \text{ W/m}\cdot\text{K}$, and Prandtl number (Pr_l) of 64. The dynamic viscosity of the liquid silicone evaluated at the tube surface temperature (μ_s) is $39.8 \times 10^{-4} \text{ kg/m}\cdot\text{s}$. The gas phase consists of air with dynamic viscosity (μ_g) of $18.4 \times 10^{-6} \text{ kg/m}\cdot\text{s}$, density (ρ_g) of 1.19 kg/m^3 , and Prandtl number (Pr_g) of 0.71. The two-phase flow has a flow quality (x) of 2.08×10^{-5} , void fraction (α) of 0.011, gas velocity (V_g) of 13.5 m/s, liquid velocity (V_l) of 9.34 m/s, and liquid mass flow rate ($\dot{m}_l$) of 0.907 kg/s. Moist air at a saturation temperature of 40°C condenses on the outer surface of the tube. The tube has a uniform surface temperature of 20°C. Determine the overall convection heat transfer coefficient (inside and outside) of the tube.

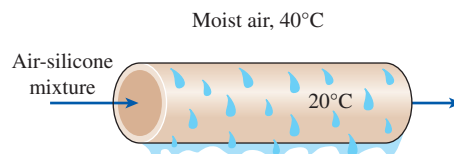


FIGURE P10-104

10-105 An air–water mixture is flowing in a 5° inclined tube that has a diameter of 25.4 mm. The two-phase mixture enters the tube at 25°C and exits at 65°C, while the tube surface

temperature is maintained at 80°C . If the superficial gas and liquid velocities are 1 m/s and 2 m/s, respectively, determine the two-phase heat transfer coefficient h_{tp} . Assume the surface tension is $\sigma = 0.068 \text{ N/m}$ and the void fraction is $\alpha = 0.33$.

10–106 A mixture of petroleum and natural gas is being transported in a pipeline with a diameter of 102 mm. The pipeline is located in a terrain that caused it to have an average inclination angle of $\theta = 10^\circ$. The liquid phase consists of petroleum with dynamic viscosity of $\mu_l = 297.5 \times 10^{-4} \text{ kg/m}\cdot\text{s}$, density of $\rho_l = 853 \text{ kg/m}^3$, thermal conductivity of $k_l = 0.163 \text{ W/m}\cdot\text{K}$, surface tension of $\sigma = 0.020 \text{ N/m}$, and Prandtl number of $\text{Pr}_l = 405$. The gas phase consists of natural gas with dynamic viscosity of $\mu_g = 9.225 \times 10^{-6} \text{ kg/m}\cdot\text{s}$, density of $\rho_g = 9.0 \text{ kg/m}^3$, and Prandtl number of $\text{Pr}_g = 0.80$. The liquid is flowing at a flow rate of 16 kg/s, while the gas is flowing at 0.055 kg/s. If the void fraction is $\alpha = 0.22$, determine the two-phase heat transfer coefficient h_{tp} . Assume the dynamic viscosity of liquid petroleum evaluated at the tube surface temperature to be $238 \times 10^{-4} \text{ kg/m}\cdot\text{s}$.

10–107 An air–water mixture is flowing in a tube with a diameter of 25.4 mm. The two-phase mixture enters the tube at 15°C and exits at 75°C , while the tube surface temperature is maintained at 80°C . If the superficial gas and liquid velocities are 1 m/s and 2 m/s, respectively, determine the two-phase heat transfer coefficient h_{tp} for (a) the horizontal tube ($\theta = 0^\circ$) and (b) the vertical tube ($\theta = 90^\circ$), and compare the two results. Assume the surface tension is $\sigma = 0.068 \text{ N/m}$ and the void fraction is $\alpha = 0.33$.

10–108 Consider a two-phase flow of air–water in a vertical upward stainless steel pipe with an inside diameter of 0.0254 m. The two-phase mixture enters the pipe at 25°C at a system pressure of 201 kPa. The superficial velocities of the water and air are 0.3 m/s and 23 m/s, respectively. The differential pressure transducer connected across the pressure taps set 1 m apart records a pressure drop of 2700 Pa, and the measured value of the void fraction is 0.86. Using the concept of the Reynolds analogy, determine the two-phase convective heat transfer coefficient. Use the following thermophysical properties for water and air: $\rho_l = 997.1 \text{ kg/m}^3$, $\mu_l = 8.9 \times 10^{-4} \text{ kg/m}\cdot\text{s}$, $\mu_s = 4.66 \times 10^{-4} \text{ kg/m}\cdot\text{s}$, $\text{Pr}_l = 6.26$, $k_l = 0.595 \text{ W/m}\cdot\text{K}$, $\sigma = 0.0719 \text{ N/m}$, $\rho_g = 2.35 \text{ kg/m}^3$, and $\mu_g = 1.84 \times 10^{-5} \text{ kg/m}\cdot\text{s}$.

Review Problems

10–109 Water is boiled at 120°C in a mechanically polished stainless steel pressure cooker placed on top of a heating unit. The inner surface of the bottom of the cooker is maintained at 130°C . The cooker that has a diameter of 20 cm and a height of 30 cm is half filled with water. Determine the time it will take for the tank to empty. *Answer: 22.8 min*

10–110 Water is to be boiled at sea level in a 30-cm-diameter mechanically polished AISI 304 stainless steel pan placed on top of a 3-kW electric burner. If 60 percent of the heat

generated by the burner is transferred to the water during boiling, determine the temperature of the inner surface of the bottom of the pan. Also, determine the temperature difference between the inner and outer surfaces of the bottom of the pan if it is 6 mm thick. Assume the boiling regime is nucleate boiling. Is this a good assumption?

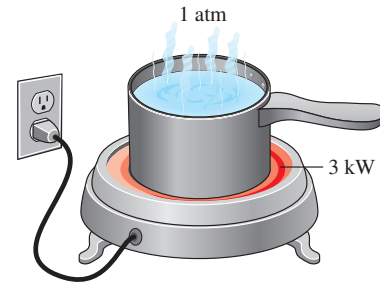


FIGURE P10–110

10–111 Repeat Prob. 10–110 for a location at an elevation of 1500 m where the atmospheric pressure is 84.5 kPa and thus the boiling temperature of water is 95°C .

10–112 A resistance heater made of 2-mm-diameter nickel wire is used to heat water at 1 atm pressure. Determine the highest temperature at which this heater can operate safely without the danger of burning out.

10–113 Water is boiled at 100°C electrically by an 80-cm-long, 2-mm-diameter horizontal resistance wire made of chemically etched stainless steel. Determine (a) the rate of heat transfer to the water and the rate of evaporation of water if the temperature of the wire is 115°C and (b) the maximum rate of evaporation in the nucleate boiling regime. *Answers: (a) 2387 W, 3.81 kg/h, (b) 1280 kW/m²*

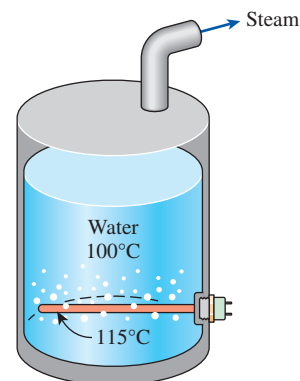


FIGURE P10–113

10–114 During a heat treatment process, a 20-mm-diameter cylindrical steel rod with a length of 200 mm was removed from a furnace when it reached a uniform temperature of

500°C and was suddenly submerged horizontally in a water bath at atmospheric pressure. If the steel rod has an emissivity of 0.9, determine the boiling heat transfer coefficient and the total heat transfer coefficient when it was initially submerged in the water bath.

10-115 Electric current is passed through a 2-mm-diameter cylindrical metal rod with emissivity of 0.5. The rod is submerged horizontally in water under atmospheric pressure, and the rod surface temperature reaches 500°C. Determine the boiling heat transfer coefficient and the total heat transfer coefficient.

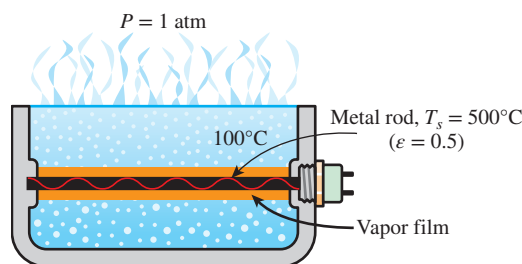


FIGURE P10-115

10-116 Water is boiled at 100°C by a spherical platinum heating element of diameter 8 cm and emissivity 0.10 immersed in the water. If the surface temperature of the heating element is 350°C, determine the convective boiling heat transfer coefficient.

10-117 In a heat treatment process, 20-mm-diameter steel ball bearings are being heated to a uniform temperature of 700°C. The ball bearings are then being removed and immediately submerged in a water bath at atmospheric pressure. If the steel ball bearings have an emissivity of 0.75, determine the boiling heat transfer coefficient and the total heat transfer coefficient when they were initially submerged in the water bath.

10-118E Saturated steam at 100°F is condensed on a 4-ft-high vertical plate that is maintained at 80°F. Determine the rate of heat transfer from the steam to the plate and the rate of condensation per foot width of the plate. Assume wavy-laminar flow. Is this a good assumption?

10-119 Saturated ammonia vapor at 25°C condenses on the outside of a 2-m-long, 3.2-cm-outer-diameter vertical tube maintained at 15°C. Determine (a) the average heat transfer coefficient, (b) the rate of heat transfer, and (c) the rate of condensation of ammonia. Assume turbulent flow and that the tube diameter is large, relative to the thickness of the liquid film at the bottom of the tube. Are these good assumptions?

10-120 Saturated refrigerant-134a vapor at 35°C is to be condensed on the outer surface of an 8-m-long, 1.2-cm-diameter

horizontal tube that is maintained at a temperature of 25°C. Determine the rate at which the refrigerant will condense in kg/min.

10-121 Repeat Prob. 10-120 for a tube diameter of 3 cm.

10-122 Steam at 40°C condenses on the outside of a 3-cm-diameter thin horizontal copper tube by cooling water that enters the tube at 10°C at an average velocity of 2 m/s and leaves at 30°C. Determine the rate of condensation of steam, the average overall heat transfer coefficient between the steam and the cooling water, and the tube length.

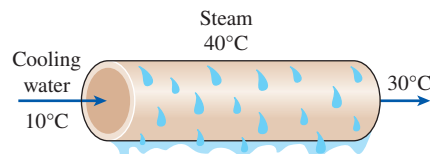


FIGURE P10-122

10-123 A 1.2-cm-diameter silver sphere initially at 25°C is suspended in a room filled with saturated steam at 100°C. Using the lumped system analysis, determine how long it will take for the temperature of the ball to rise to 50°C. Also determine the amount of steam that condenses during this process, and verify that the lumped system analysis is applicable.

10-124 Repeat Prob. 10-123 for a 3-cm-diameter copper ball.

10-125 Saturated isobutane vapor in a binary geothermal power plant is to be condensed outside an array of eight horizontal tubes. Determine the ratio of the condensation rate for the cases of the tubes being arranged in a horizontal tier versus in a vertical tier of horizontal tubes.

10-126E The condenser of a steam power plant operates at a pressure of 0.95 psia. The condenser consists of 100 horizontal tubes arranged in a 10 × 10 square array. The tubes are 15 ft long and have an outer diameter of 1.2 in. If the outer surfaces of the tubes are maintained at 80°F, determine (a) the rate of heat transfer from the steam to the cooling water and (b) the rate of condensation of steam in the condenser.

10-127E Repeat Prob. 10-126E for a tube diameter of 2 in.

10-128 Shown in Fig. P10-128 is the tube layout for a horizontal condenser that is used for liquefying 900 kg/h of saturated ammonia vapor at 37°C. There are 14 copper tubes, each with inner diameter $D_i = 3.0$ cm and outer diameter $D_o = 3.8$ cm. A coolant flows through the tubes at an average temperature of 20°C such that it yields a heat transfer coefficient of 4.0 kW/m²·K. For this condenser, estimate (a) the average value of the overall heat transfer coefficient and (b) the tube length. Assume a tube outer surface temperature of 31°C. Is this a good assumption?

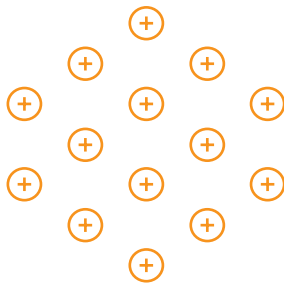


FIGURE P10-128

10-129 Saturated ammonia vapor at 25°C condenses on the outside surface of 16 thin-walled tubes, 2.5 cm in diameter, arranged horizontally in a 4 × 4 square array. Cooling water enters the tubes at 14°C at an average velocity of 2 m/s and exits at 17°C. Calculate (a) the rate of NH₃ condensation, (b) the overall heat transfer coefficient, and (c) the tube length.

10-130 Saturated steam at 270.1 kPa condenses inside a horizontal, 10-m-long, 2.5-cm-internal-diameter pipe whose surface is maintained at 110°C. Assuming low vapor velocity, determine the average heat transfer coefficient and the rate of condensation of the steam inside the pipe. *Answers: 8413 W/m²·K, 0.0608 kg/s*

Fundamentals of Engineering (FE) Exam Problems

10-131 When boiling a saturated liquid, one must be careful while increasing the heat flux to avoid burnout. Burnout occurs when the boiling transitions from _____ boiling.

- (a) convection to nucleate
- (b) convection to film
- (c) film to nucleate
- (d) nucleate to film
- (e) none of them

10-132 Heat transfer coefficients for a vapor condensing on a surface can be increased by promoting

- (a) film condensation
- (b) dropwise condensation
- (c) rolling action
- (d) none of them

10-133 At a distance x down a vertical, isothermal flat plate on which a saturated vapor is condensing in a continuous film, the thickness of the liquid condensate layer is δ . The heat transfer coefficient at this location on the plate is given by

- (a) k_l/δ
- (b) δh_f
- (c) δh_{fg}
- (d) δh_g
- (e) none of them

10-134 When a saturated vapor condenses on a vertical, isothermal flat plate in a continuous film, the rate of heat transfer is proportional to

- (a) $(T_s - T_{\text{sat}})^{1/4}$
- (b) $(T_s - T_{\text{sat}})^{1/2}$
- (c) $(T_s - T_{\text{sat}})^{3/4}$
- (d) $(T_s - T_{\text{sat}})$
- (e) $(T_s - T_{\text{sat}})^{2/3}$

10-135 Saturated water vapor is condensing on a 0.5 m² vertical flat plate in a continuous film with an average heat transfer coefficient of 5 kW/m²·K. The temperature of the water is 80°C ($h_{fg} = 2309$ kJ/kg), and the temperature of the plate is 60°C. The rate at which condensate is being formed is

- (a) 0.022 kg/s
- (b) 0.048 kg/s
- (c) 0.077 kg/s
- (d) 0.16 kg/s
- (e) 0.32 kg/s

10-136 Steam condenses at 50°C on a 1.2-m-high and 2.4-m-wide vertical plate that is maintained at 30°C. The condensation heat transfer coefficient is

- (a) 4260 W/m²·K
- (b) 4780 W/m²·K
- (c) 5510 W/m²·K
- (d) 6260 W/m²·K
- (e) 6940 W/m²·K

(For water, use $\rho_l = 992.1$ kg/m³, $\mu_l = 0.653 \times 10^{-3}$ kg/m·s, $k_l = 0.631$ W/m·K, $c_{pl} = 4179$ J/kg·°C, $h_{fg@T_{\text{sat}}} = 2383$ kJ/kg)

10-137 An air conditioner condenser in an automobile consists of 2 m² of tubular heat exchange area whose surface temperature is 30°C. Saturated refrigerant-134a vapor at 50°C ($h_{fg} = 152$ kJ/kg) condenses on these tubes. What heat transfer coefficient must exist between the tube surface and condensing vapor to produce 1.5 kg/min of condensate?

- (a) 95 W/m²·K
- (b) 640 W/m²·K
- (c) 727 W/m²·K
- (d) 799 W/m²·K
- (e) 960 W/m²·K

10-138 Saturated water vapor at 40°C is to be condensed as it flows through a tube at a rate of 0.2 kg/s. The condensate leaves the tube as a saturated liquid at 40°C. The rate of heat transfer from the tube is

- (a) 34 kJ/s
- (b) 268 kJ/s
- (c) 453 kJ/s
- (d) 481 kJ/s
- (e) 515 kJ/s

10-139 Steam condenses at 50°C on the outer surface of a horizontal tube with an outer diameter of 6 cm. The outer surface of the tube is maintained at 30°C. The condensation heat transfer coefficient is

- (a) 5493 W/m²·K
- (b) 5921 W/m²·K
- (c) 6796 W/m²·K
- (d) 7040 W/m²·K
- (e) 7350 W/m²·K

(For water, use $\rho_l = 992.1$ kg/m³, $\mu_l = 0.653 \times 10^{-3}$ kg/m·s, $k_l = 0.631$ W/m·K, $c_{pl} = 4179$ J/kg·°C, $h_{fg@T_{\text{sat}}} = 2383$ kJ/kg)

10-140 Steam condenses at 50°C on the tube bank consisting of 20 tubes arranged in a rectangular array of 4 tubes high and 5 tubes wide. Each tube has a diameter of 3 cm and a length of 5 m, and the outer surfaces of the tubes are maintained at 30°C. The rate of condensation of steam is

- (a) 0.12 kg/s
- (b) 0.28 kg/s
- (c) 0.31 kg/s
- (d) 0.45 kg/s
- (e) 0.62 kg/s

(For water, use $\rho_l = 992.1$ kg/m³, $\mu_l = 0.653 \times 10^{-3}$ kg/m·s, $k_l = 0.631$ W/m·K, $c_{pl} = 4179$ J/kg·°C, $h_{fg@T_{\text{sat}}} = 2383$ kJ/kg)

Design and Essay Problems

10–141 Design the condenser of a steam power plant that has a thermal efficiency of 40 percent and generates 10 MW of net electric power. Steam enters the condenser as saturated vapor at 10 kPa, and it is to be condensed outside horizontal tubes through which cooling water from a nearby river flows. The temperature rise of the cooling water is limited to 8°C, and the velocity of the cooling water in the pipes is limited to 6 m/s to keep the pressure drop at an acceptable level. Specify the pipe diameter, the total pipe length, and the arrangement of the pipes to minimize the condenser volume.

10–142 The refrigerant in a household refrigerator is condensed as it flows through the coil that is typically placed behind the refrigerator. Heat transfer from the outer surface of the coil to the surroundings is by natural convection and radiation. Obtaining information about the operating conditions of the refrigerator, including the pressures and temperatures of the refrigerant at the inlet and the exit of the coil, show that the coil is selected properly, and determine the safety margin in the selection.

10–143 Water-cooled steam condensers are commonly used in steam power plants. Obtain information about water-cooled steam condensers by doing a literature search on the topic and also by contacting some condenser manufacturers. In a report, describe the various types, the way they are designed, the limitation on each type, and the selection criteria.

10–144 A manufacturing facility requires saturated steam at 120°C at a rate of 1.2 kg/min. Design an electric steam boiler for this purpose under these constraints:

- The boiler will be cylindrical with a height-to-diameter ratio of 1.5. The boiler can be horizontal or vertical.
- The boiler will operate in the nucleate boiling regime, and the design heat flux will not exceed 60 percent of the critical heat flux to provide an adequate safety margin.

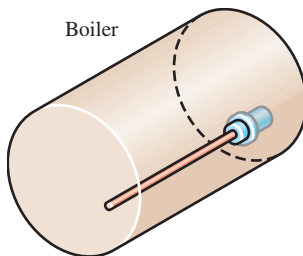


FIGURE P10–144

- A commercially available plug-in-type electrical heating element made of mechanically polished stainless steel will be used. The diameter of the heater cannot be between 0.5 cm and 3 cm.
- Half of the volume of the boiler should be occupied by steam, and the boiler should be large enough to hold enough water for a 2-h supply of steam. Also, the boiler will be well insulated.

You are to specify the following: (a) The height and inner diameter of the tank, (b) the length, diameter, power rating, and surface temperature of the electric heating element, (c) the maximum rate of steam production during short periods of overload conditions, and how it can be accomplished.

10–145 Repeat Prob. 10–144 for a boiler that produces steam at 150°C at a rate of 2.5 kg/min.

10–146 Conduct this experiment to determine the boiling heat transfer coefficient. You will need a portable immersion-type electric heating element, an indoor-outdoor thermometer, and metal glue (all can be purchased for about \$15 in a hardware store). You will also need a piece of string and a ruler to calculate the surface area of the heater. First, boil water in a pan using the heating element and measure the temperature of the boiling water away from the heating element. Based on your reading, estimate the elevation of your location, and compare it to the actual value. Then glue the tip of the thermocouple wire of the thermometer to the midsection of the heater surface. The temperature reading in this case will give the surface temperature of the heater. Assuming the rated power of the heater to be the actual power consumption during heating (you can check this by measuring the electric current and voltage), calculate the heat transfer coefficients from Newton's law of cooling.

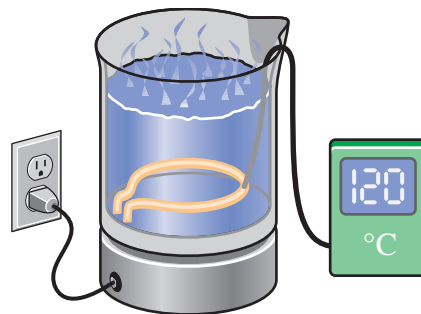


FIGURE P10–146